Defective Analysis

A Unified Internal Treatment

Framing, Defective Calculus, a Graph Worked Example, the Scalar Ladder, Geometry, COTG, and a Physics Schema

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Postulated analysis needing confirmation and review

Editorial preface to the consolidated edition

This edition merges eight working documents into one spine: the framing and terminology of Defective Analysis (DA); Defective Calculus (DC) as its formal subcalculus; Kelly's lemma as a worked example; the generator-presentation derivation of the scalar ladder $\mathbb{N} \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{R} \to \mathbb{C}$; a formal exploration of geometry; Charted Oscillatory Transport Geometry (COTG) derived through the Whole Defect System (WDS) path; a global-definitions addendum; and a physics-unification schema. It is an internal working reference, not a publication.

Corrections applied. The source documents were reviewed line by line. The corrections folded into this edition are recorded in Appendix A with a one-line rationale each. The most consequential were: repairing the circular statement of no-leakage in DC so that it constrains downstream visible composites rather than restating a tautology; fixing the domain of the polarity classifier σ to a single signature ($\sigma : Y \to \Sigma \cup \{\perp\}$) used uniformly; separating the deck *fiber* from the retained *kernel* in the Kelly example, where the kernel is selection data and the fiber carries the candidate class; restoring transitivity to reconstruction equivalence; typing the clock-order residual into a single residual carrier before any commutator shorthand is written; and demoting several framework restatements from "Theorem" to "Proposition" or "Remark" so that the word *theorem* continues to mark genuine deductions.

Four design decisions fix conventions that the source documents left oscillating. They are stated once here and held throughout.

(D1) **Retained kernel = selection data.** For $X \in \mathcal{X}$, the fiber $F_\theta(D_\theta X)$ carries the candidate class; the retained kernel $K_{\mathrm{ret}}(X)$ is the admissible data one must be *handed* to land on X inside that fiber. The two are never identified, even when a fiber happens to be a singleton.

(D2) $\perp$ **is structural.** $\sigma(y) = \perp$ asserts that, relative to the *declared* instance, no legal operation resolves the polarity over y. It is a property of the instance, not a confession of ignorance; enlarging the declared legal operations is a change of instance, not a resolution of the same one.

(D3) **The clock-order residual is fundamental.** Noncommutativity of D_θ and I_α is taken as primary structure, not an artifact of flat reconstruction. Closure of a diagram under *complete* reconstruction is a per-instance theorem target, to be proved where it holds, never presumed.

(D4) **Clock-index doctrine.** A clock is an index over admissible operations; ordinary time is a (different) index; a coordinate/dimension is a (third) index. *Clock $\neq$ time $\neq$ dimension.*

> In the scalar ladder the symbols θ, α are clocks in exactly this sense—indices over the additive/successor and multiplicative/aggregation operations—so they keep their names there as in the spine. The bridge is stated explicitly in §4.1 rather than left implicit.

Status of claims. Every chapter retains its own claim boundary. Nothing here is asserted as an experimentally confirmed theory, a completed derivation of quantum mechanics or general relativity, or a solution to any open problem in standard mathematics. The graph chapter proves Kelly's lemma and nothing more about reconstruction; the scalar-ladder chapter proves what it states up to one clearly marked terminal postulate; the physics chapter is explicitly a typed schema with its open obligations preserved.

Contents

Chapter 1

Framing, Terminology, and Problem Description

Claim boundary. This chapter defines a proposed analytic vocabulary in ordinary mathematical language: set, class, map, equivalence relation, projection, fiber, kernel, quotient, section, residual, admissibility, and clock signature. The symbols θ and α are local notation for two operation modes; they are not asserted to be new mathematical structures. The local/global classifier is a postulated diagnostic rule, not a completed theorem. No claim is made about external applications, physical systems, or open mathematical conjectures.

Defective Analysis (DA) describes a problem by assigning an operation clock, identifying the data exposed by that clock, and stating what remains hidden in the corresponding fiber. The first diagnostic task is clock assignment. A problem is *locally determined* when it is governed by exactly one clock—either a θ-clock or an α-clock. It is *globally coupled* when both clocks are required. It is *malformed* when the assigned clock signature is wrong. This chapter fixes the basic terminology and gives a protocol for describing a problem without smuggling in side information.

1.1 Purpose

Defective Analysis is the top-level framework. It contains, but is not identical with, Defective Calculus, Whole Defect Systems, Scalar Calculus, and Charted Oscillatory Transport Geometry. Its purpose is to describe a problem before attempting to solve it. The basic DA question is not immediately "what is the answer?" It is: *which clock structure makes the problem well-formed?* Once the clock structure is assigned, DA asks what a legal projection exposes, what the corresponding reconstruction must retain, and whether the remaining ambiguity is admissible or the result of a wrong assignment.

1.2 Primitive terms

A DA description of a problem begins with the following data.

Symbol	Meaning
P	The problem being described.
$\mathcal{X}$	A carrier class of admissible objects or states.
$\sim$	The declared equivalence relation on candidate solutions.
θ	The derivation / projection / probe clock.
α	The integration / realization / reconstruction clock.
D_θ	The defective observable generated by the θ-clock.
I_α	The reconstruction operation generated by the α-clock.
F_θ	The fiber of states sharing the same defective observable.
K_{ret}	The admissible retained-kernel (selection-data) class.
Π	The visible quotient or visible-data projection.
σ	The polarity classifier; $\sigma = \perp$ means unresolved polarity.
Adm	The admissibility rule preventing leakage and invalid reconstruction.

A clock is not a literal time variable; by the clock-index doctrine (D4) it is an index over an allowed mode of operation. A θ-clock exposes, deletes, differentiates, probes, projects, or coarse-grains. An α-clock glues, integrates, completes, realizes, evolves, or reconstructs.

1.3 Clock assignment

Definition 1.1 (Clock signature)**.** The clock signature of a problem P is a subset $\mathrm{Clk}(P) \subseteq \{\theta, \alpha\}$ recording which operation clocks are necessary for P to be well-formed.

Postulate 1.2 (Local/global diagnostic)**.** The first DA diagnostic is the clock-classification rule

$$\text{local} \;=\; \theta \veebar \alpha, \qquad \text{global} \;=\; \theta \wedge \alpha,$$

with a third, *type-distinct*, failure verdict: P is *malformed* when the assigned signature does not match the operations that actually make P well-formed.

Here "local" means single-clock deterministic; it does not mean spatially small. A local problem is governed by exactly one clock. A global problem requires both clocks and therefore their interaction, order, compatibility, or closure.

[Correction: "malformed" is a predicate on the assignment $\mathrm{Clk}_{assigned}(P) \neq \mathrm{Clk}_{true}(P)$*, not a third element of the subset* $\mathrm{Clk}(P) \subseteq \{\theta, \alpha\}$*. The source document's four-case "*$\mathrm{Class}_{\mathrm{DA}}$*" mixed a value of* Clk *with a property of the assignment; the two-axis form below keeps the types separate.]*

DA thus reports a pair: a well-formed signature when the assignment is correct,

$$\mathrm{Clk}(P) \in \big\{\{\theta\}, \{\alpha\}, \{\theta, \alpha\}\big\},$$

together with the verdict WELL-ASSIGNED or MALFORMED according to whether $\mathrm{Clk}_{\text{assigned}}(P) = \mathrm{Clk}_{\text{true}}(P)$.

1.4 Problem-description protocol

A DA problem description states the following before any attempted solution.

1. **Carrier.** Declare the object class $\mathcal{X}$.
2. **Equivalence.** Declare $\sim$, the relation under which two solutions count as the same.
3. **Given data.** Declare what data are actually supplied.
4. **Legal operations.** Declare the legal θ- and α-operations.
5. **Clock signature.** State whether P is θ-local, α-local, $\theta\alpha$-global, or currently malformed.
6. **Projection.** If present, define $D_\theta : \mathcal{X} \to Y$.
7. **Fiber.** Define the unresolved fiber $D_\theta^{-1}(D_\theta X)$.
8. **Retained kernel.** State what class of hidden *selection data* is admissible to retain.
9. **Polarity.** State whether the polarity is determined or unresolved.
10. **No-leakage rule.** State what information cannot be introduced without invalidating the problem.

This protocol keeps the framework reviewable. A problem may be expressive and still invalid if its solution uses data that were not supplied or legally generated.

1.5 Local, global, and malformed problems

A θ-local problem is deterministic under projection, differentiation, deletion, or probing alone. An α-local problem is deterministic under reconstruction, completion, gluing, or realization alone. A $\theta\alpha$-global problem is one in which the interaction of projection and reconstruction is essential. A typical global obstruction has the form

$$D_\theta I_\alpha \;\neq\; I_\alpha D_\theta$$

when both composites are meaningful (the precise typing of this comparison is fixed in Chapter 2 and Chapter 7; by (D3) the residual is taken as fundamental). The failure of these operations to close is the *defect residual*. The central failure mode is not mystery; it is wrong assignment, $\mathrm{Clk}_{\mathrm{assigned}}(P) \neq \mathrm{Clk}_{\mathrm{true}}(P)$, which can create artificial defects. DA therefore requires clock diagnosis before interpreting an obstruction as intrinsic.

1.6 Static and operational regimes

Definition 1.3 (Static regime)**.** In a static regime, the available information is only the declared observable $D_\theta(\mathcal{X})$. A scalar response, operator action, or hidden support may be used only if it is already encoded in the declared observable. Otherwise it is leakage.

Definition 1.4 (Operational regime)**.** In an operational regime, the object is hidden but can be queried through declared measurement ports and legal clock-generated operations. A probe is legal only if it is produced from an allowed port through an admissible $\langle \alpha, \theta \rangle$ sector. If P is the declared measurement-port subspace and $O_{\alpha,\theta} = \langle \alpha, \theta \rangle$ is the legal operation algebra, then the legal probe space is $Z_{\mathrm{legal}} = \mathrm{Sector}_{\alpha,\theta}(P)$.

In an operational regime one may measure scalar responses of legal probes. In a static regime the same response is inadmissible unless it is part of the data.

1.7 No-leakage

Definition 1.5 (No-leakage, framing level)**.** A DA argument satisfies no-leakage when every quantity used in the solution is either contained in the stated data or generated by a declared legal operation. If a quantity identifies the hidden representative, hidden support, or missing reconstruction choice without being legal data, the argument leaks.

No-leakage is the discipline that distinguishes a legitimate retained kernel from an arbitrary rescue channel. Its precise form—a constraint on downstream visible composites—is given in Chapter 2.

1.8 Relation to subordinate chapters

Defective Calculus (Chapter 2) is the subcalculus of D_θ, I_α, fibers, retained kernels, residuals, and admissibility. A Whole Defect System (Chapters 6–7) is an upstream object in which a carrier is equipped with a defect extractor, defect sectors, a gluing carrier, and a quotient/gluing projection. COTG (Chapter 6) is a charted oscillatory realization layer derived through a WDS path. Scalar Calculus (Chapter 7) is a finite quotient and parity-kernel model used to test that the grammar is nonempty, constrained, and falsifiable.

1.9 Failure surfaces

Code	Failure surface
F1	Wrong clock assignment: P treated as θ-local, α-local, or global incorrectly.
F2	Leakage: the solution uses data not supplied or legally generated.
F3	Undefined equivalence: the relation $\sim$ is not stated.
F4	Arbitrary retained kernel: hidden data added without an admissibility rule.
F5	Projection/reconstruction mismatch: D_θ and I_α act in incompatible carriers.
F6	False polarity: $\sigma = \perp$ collapsed without a legal resolving operation.
F7	Static/operational conflation: a probe response used in a static problem without a port.

1.10 Known unknowns and proof obligations

1. Define categories of DA problem descriptions and morphisms between them.
2. Specify when the true clock signature can be derived rather than assigned.
3. State conditions under which polarity is derived from a fiber and when it must be primitive. (By (D2) the present edition reads $\perp$ as structural; a derivation criterion would refine this.)
4. Formalize legal probe generation in operational regimes without smuggling hidden representatives.
5. Relate DA failure surfaces to standard mathematical counterexamples and collapse tests.

1.11 Final compression

Defective Analysis assigns clock signatures to problems and then studies the observable, fiber, retained kernel, polarity, and admissibility induced by that assignment:

$$\mathrm{DA}(P) = \big(\mathrm{Clk}(P),\, D_\theta,\, I_\alpha,\, F_\theta,\, K_{\mathrm{ret}},\, \Pi,\, \sigma,\, \mathrm{Adm}\big),$$

$$\text{local} = \theta \veebar \alpha, \qquad \text{global} = \theta \wedge \alpha, \qquad \text{failure} = \text{wrong clock assignment}.$$

Chapter 2

Defective Calculus

Claim boundary. This chapter defines Defective Calculus as a proposed formal subcalculus of Defective Analysis, in ordinary set-theoretic and categorical language. The retained kernel is an admissible class of *selection data* (decision D1), not arbitrary missing information. Polarity is taken as structural (decision D2). This chapter states a grammar for review, not a completed universal theorem.

Defective Calculus studies clocked projections and admissible reconstructions. Given a carrier $\mathcal{X}$, a defective observable $D_\theta : \mathcal{X} \to Y$, and a reconstruction operation I_α, the central object is the fiber of states sharing the same observable. Reconstruction succeeds only when the retained kernel is supplied through an admissible rule. The calculus records flat reconstruction, complete reconstruction, residuals, no-leakage, and polarity collapse.

2.1 Position inside DA

Defective Analysis first assigns the clock signature of a problem. Defective Calculus begins once a projection/reconstruction structure has been declared. It studies the pattern

$$\mathcal{X} \xrightarrow{D_\theta} Y \qquad \text{and} \qquad (Y, k) \xrightarrow{I_\alpha} \mathcal{X}/{\sim},$$

where D_θ is the defective observable and I_α is reconstruction using retained data k.

2.2 Defective-calculus instance

Definition 2.1 (Defective-calculus instance)**.** A defective-calculus instance is a tuple

$$\mathcal{C} = (\mathcal{X},\ \sim,\ Y,\ K,\ D_\theta,\ I_\alpha,\ \Pi,\ \sigma,\ \mathrm{Adm})$$

where $\mathcal{X}$ is a carrier class; $\sim$ is the declared reconstruction equivalence; Y is the observable-data class; K is the candidate retained-data class; $D_\theta : \mathcal{X} \to Y$ is the defective observable; $I_\alpha : Y \times K \to \mathcal{X}/{\sim}$ is reconstruction modulo $\sim$; Π is a visible quotient or comparison map; σ is a polarity classifier (Definition 2.8); and Adm is the admissibility predicate.

The same data may be read as set/fiber language (emphasizing preimages) or as category language (emphasizing maps, quotient objects, and sections).

2.3 Fibers and reconstruction

Definition 2.2 (Defective fiber)**.** For $y \in Y$, the θ-fiber over y is $F_\theta(y) = D_\theta^{-1}(y) = \{X \in \mathcal{X} : D_\theta X = y\}$. For $X \in \mathcal{X}$, the unresolved reconstruction fiber is $F_\theta(X) = D_\theta^{-1}(D_\theta X)$.

By decision (D1) this fiber is the carrier of the candidate class; it is *not* the retained kernel, which is defined in §2.4 as the selection data needed to single out a representative within it.

Definition 2.3 (Flat reconstruction)**.** If $0 \in K$ is a declared null retained datum, the flat reconstruction is $I_\alpha^0 D_\theta(X) = I_\alpha(D_\theta X, 0)$. Flat reconstruction is the reconstruction produced by visible data alone.

Definition 2.4 (Defect)**.** An object X has nontrivial defect relative to (D_θ, I_α) when $I_\alpha^0 D_\theta(X) \not\sim X$. The defect is not necessarily an error; it is the obstruction exposed by the declared projection/reconstruction structure.

2.4 Retained kernel as admissible selection data

Definition 2.5 (Retained-kernel class)**.** For $X \in \mathcal{X}$, the retained-kernel class is

$$K_{\mathrm{ret}}(X) = \{\, k \in K : I_\alpha(D_\theta X, k) \sim X \text{ and } \mathrm{Adm}(X, k) \,\}.$$

Thus $K_{\mathrm{ret}}(X)$ is not arbitrary missing information; it is the admissible class of *selection data* that, supplied alongside the visible observable, reconstructs X up to $\sim$. A complete reconstruction is admissible when there exists $k \in K_{\mathrm{ret}}(X)$ with $I_\alpha(D_\theta X, k) \sim X$. The retained kernel may be a single datum, a finite set, a section choice, a parity class, or a constrained probe response, depending on the realization; its admissibility must be stated.

2.5 Visible quotient and no-leakage

Definition 2.6 (Visible quotient and hidden sector)**.** Let $\Pi : K \to Q$ be a visible quotient on retained data. The hidden sector is $K_{\mathrm{hid}} = \{k \in K : \Pi(k) = 0\}$ when Q has a distinguished zero; otherwise K_{hid} is the equivalence induced by equality under Π. Write $k \sim_{\mathrm{hid}} k'$ for $\Pi(k) = \Pi(k')$.

Definition 2.7 (No-leakage)**.** No-leakage holds when hidden-equivalent retained data are indistinguishable to every declared visible operation built from the instance. Concretely, for all $y \in Y$ and $k, k' \in K$,

$$k \sim_{\mathrm{hid}} k' \implies D_\theta\big(I_\alpha(y, k)\big) = D_\theta\big(I_\alpha(y, k')\big) \text{ and } \Pi(k) = \Pi(k'),$$

whenever D_θ descends to $\mathcal{X}/\sim$ on the relevant classes. Hidden data may be retained, but data killed by the quotient may not reappear as visible certainty.

[Correction E1: the source statement "$k \sim_{K_{\mathrm{hid}}} k' \Rightarrow \Pi(k) = \Pi(k')$" is a tautology under $K_{\mathrm{hid}} = \ker \Pi$ and carries no content. The intended discipline is a constraint on downstream *visible composites: hidden-equivalent retained data must produce reconstructions that no legal visible map can separate. The display above states exactly that. A bad retained kernel violates no-leakage when two hidden-equivalent data yield reconstructions a declared visible operation can tell apart.]*

2.6 Polarity

Definition 2.8 (Polarity classifier)**.** A polarity classifier is a map

$$\sigma : Y \longrightarrow \Sigma \cup \{\bot\},$$

where Σ is a declared set of determined polarity classes and $\bot$ denotes unresolved polarity. The value $\sigma(y)$ is a property of the fiber $F_\theta(y)$. An extended classifier $\sigma(y, \rho)$ may take additional probe data ρ in the operational regime.

[Correction E2: the source gave σ three incompatible domains (fiber elements; $D_\theta X \in Y$; and $(D_\theta X, \text{probe})$). Fixing the single signature $\sigma : Y \to \Sigma \cup \{\bot\}$, with an explicit extended form for the operational regime, removes the ambiguity and matches the usage $\sigma(\mathrm{Deck}(G))$ in Chapter 3.]

By decision (D2), $\sigma(y) = \bot$ asserts that the declared instance admits no legal operation determining which hidden class over y is active. It is not a license to choose a representative without additional legal information; in the operational regime, only an admissible probe (not an appeal to ignorance) can move σ off $\bot$, and doing so is governed by the extended classifier.

2.7 Residuals and clock order

When both clocks are active, the central residual is the failure of projection and reconstruction to close. The residual depends on the retained datum used, so we carry k explicitly. In an additive or comparable residual carrier R,

$$\Delta_{\theta,\alpha}(X; k) \;=\; D_\theta\, I_\alpha(D_\theta X, k) \;-\; D_\theta X.$$

More generally $\Delta_{\theta,\alpha}(\,\cdot\,; k) : \mathcal{X} \to R$ maps into a declared residual carrier. A zero residual, $\Delta_{\theta,\alpha}(X; k) = 0$, means the relevant diagram closes for that datum; a nonzero residual records the obstruction produced by clock interaction.

[Correction E3: the residual is written $\Delta_{\theta,\alpha}(X; k)$ to expose its dependence on the retained datum k, which the source suppressed. Both terms live in Y (then mapped to R), so the comparison is well-typed without requiring D_θ to act on a reconstructed class; cf. Correction E10 in Chapter 7, where the source's $D_\theta I_\alpha$ vs $I_\alpha D_\theta$ form is repaired.]

2.8 Sections and representative choice

If $D_\theta : \mathcal{X} \to \mathrm{im}(D_\theta)$ is surjective onto its image, a section is a map $s : \mathrm{im}(D_\theta) \to \mathcal{X}$ with $D_\theta \circ s = \mathrm{id}_{\mathrm{im}(D_\theta)}$. The opposite composite $s \circ D_\theta : \mathcal{X} \to \mathcal{X}$ is a representative-selection operator. It equals the identity only when every fiber is a singleton or when the section is supplied with sufficient representative-specifying data—that is, with an element of the retained kernel.

2.9 Operational probes

In an operational regime, a hidden system may be queried through declared measurement ports. Let A be a declared internal transport operator and P the legal measurement-port subspace, so the legal probe space is $Z_{\text{legal}} = \text{Sector}_{\alpha,\theta}(P)$. For $z \in Z_{\text{legal}}$, the scalar return response at step m is

$$R_z(m) = z^* A^m z, \qquad m = 0, 1, 2, \ldots$$

This response is legal only in the operational regime; in a static regime it is leakage unless $R_z(m)$ is already contained in the supplied observable. A minimal legal probe solves

$$z^* = \arg\min_{z \in Z_{\text{legal}}} \text{Cost}(z), \qquad \text{subject to } \exists M : \ \sigma(D_\theta X) = \perp, \ \sigma\big(D_\theta X, \ R_z[0{:}M]\big) \neq \perp.$$

The hidden support is not directly queried; it may only be inferred from legal scalar responses.

[Correction E4: the source reused k for both the operator power and the retained datum, and K for both the integer cutoff and the retained-data class. Here the probe step is m, the cutoff is M, and K is reserved for the class.]

2.10 Admissibility

Definition 2.9 (Admissible reconstruction)**.** A reconstruction is admissible when all of the following hold: (1) the observable was produced by a declared D_θ; (2) the reconstruction uses only declared I_α operations; (3) retained data lie in K_{ret}; (4) no-leakage holds (Definition 2.7); (5) the result equals the target up to $\sim$, or else the residual is explicitly recorded.

2.11 Failure surfaces

Code	Failure surface
F1	Undefined carrier or equivalence relation.
F2	Projection exists but reconstruction target is not declared.
F3	Retained kernel is arbitrary rather than admissible selection data.
F4	Hidden differences leak into visible data (violation of Definition 2.7).
F5	$\sigma = \perp$ collapsed without legal information.
F6	Residual carrier not specified, so closure/failure cannot be tested.
F7	Static and operational regimes conflated.
F8	Probe response assumes an operator not supplied or legally queriable.

2.12 Known unknowns and proof obligations

1. Give category-level conditions under which I_α is a genuine section, adjoint, quotient inverse, or completion.
2. Characterize when σ is structurally determined and when $\perp$ is forced by the declared instance.
3. Classify admissible retained kernels in standard mathematical examples.
4. Give finite examples where no-leakage passes and nearby examples where it fails.
5. Separate static reconstruction problems from operational probe-selection problems.

2.13 Final compression

Defective Calculus is the algebra of clocked projection and admissible reconstruction:

$$\mathcal{X} \xrightarrow{D_\theta} D_\theta X, \qquad (D_\theta X, k) \xrightarrow{I_\alpha} \mathcal{X}/\!\sim,$$

$$K_{\mathrm{ret}}(X) = \{k : I_\alpha(D_\theta X, k) \sim X \text{ and } \mathrm{Adm}(X, k)\}.$$

$\sigma(y) = \perp$ means the declared instance does not determine the hidden polarity over y. No-leakage: hidden data may be retained, but not smuggled into visibility.

Chapter 3

Kelly's Lemma and Graph Reconstruction

Claim boundary. This chapter is a worked example. It proves Kelly's lemma by a standard double count and then translates the result into Defective Calculus. It does *not* prove the graph reconstruction conjecture and introduces no new graph invariant. It records how DA describes the projection, fiber, retained kernel, section, and unresolved ambiguity of the deck map.

For a finite simple graph G, the vertex-deleted deck is the multiset of cards $G - v$. Kelly's lemma states that every proper subgraph count is determined by the deck. In defective-calculus terms the deck map is a θ-projection and the inverse image of a deck is the reconstruction fiber. A section of the deck map chooses a representative; the reconstruction conjecture is the claim that the relevant deck fibers are singletons for graphs with at least three vertices.

3.1 Standard graph-theoretic setup

All graphs here are finite, simple, and undirected. For a graph G, write $V(G)$ for its vertex set and $|G| = |V(G)|$. Let U_n be the set of isomorphism classes of n-vertex finite simple graphs, and $[G] \in U_n$ the class of G.

Definition 3.1 (Vertex-deleted deck)**.** For an n-vertex graph G, its vertex-deleted deck is the multiset $\mathrm{Deck}(G) = \{[G - v] : v \in V(G)\}$, with multiplicity.

Definition 3.2 (Deck map)**.** The deck map is $\delta_n : U_n \to \mathrm{MSet}_n(U_{n-1})$, $\delta_n([G]) = \mathrm{Deck}(G)$. When convenient the codomain is restricted to $\mathrm{im}(\delta_n)$.

Definition 3.3 (Hypomorphism and deck fiber)**.** Two n-vertex graphs G, H are hypomorphic when $\mathrm{Deck}(G) = \mathrm{Deck}(H)$. For a realized deck $D \in \mathrm{im}(\delta_n)$, the deck fiber is $\mathrm{Fib}(D) = \delta_n^{-1}(D) = \{[H] \in U_n : \delta_n([H]) = D\}$.

Definition 3.4 (Reconstructible parameter)**.** A graph parameter $p : U_n \to A$ is reconstructible from the deck if there exists $\tilde{p} : \mathrm{im}(\delta_n) \to A$ with $p = \tilde{p} \circ \delta_n$; equivalently, p is constant on every deck fiber.

3.2 Kelly's lemma

Let $N(F,G)$ denote the number of subgraphs of G isomorphic to F, and $N_{\mathrm{ind}}(F,G)$ the number of induced such subgraphs.

Lemma 3.5 (Kelly)**.** *Let G be an n-vertex graph and F a graph with $m < n$ vertices. Then*

$$N(F,G) = \frac{1}{n-m} \sum_{C \in \mathrm{Deck}(G)} N(F,C),$$

summing cards with multiplicity. The same formula holds for induced counts:

$$N_{\mathrm{ind}}(F,G) = \frac{1}{n-m} \sum_{C \in \mathrm{Deck}(G)} N_{\mathrm{ind}}(F,C).$$

Proof. Consider the incidence set $\Omega = \{(v,X) : v \in V(G),\ X \subseteq G,\ X \cong F,\ v \notin V(X)\}$ and count it two ways. Fixing a copy $X \cong F$: it uses m vertices, so $n-m$ vertices lie outside X, and deleting any outside vertex leaves X visible in the card; hence $|\Omega| = (n-m)N(F,G)$. Fixing a vertex v: the copies of F avoiding v are exactly the copies of F in $G-v$, so $|\Omega| = \sum_v N(F,G-v) = \sum_{C\in\mathrm{Deck}(G)} N(F,C)$. Equating gives the formula. The induced version is identical: an induced copy on a vertex set S survives exactly in the cards obtained by deleting vertices outside S. $\square$

Corollary 3.6 (Kelly counts factor through the deck)**.** *For every F with $|F| < n$, the parameters $[G] \mapsto N(F,G)$ and $[G] \mapsto N_{\mathrm{ind}}(F,G)$ are reconstructible from the deck.*

3.3 The deck as a complete $(n-1)$-count vector

For $F \in U_{n-1}$, the multiplicity of F in $\mathrm{Deck}(G)$ is exactly $N_{\mathrm{ind}}(F,G)$. Thus the deck is equivalently the vector of all induced $(n-1)$-vertex subgraph counts, $\delta_n([G]) = \big(N_{\mathrm{ind}}(F,G)\big)_{F\in U_{n-1}}$, and Kelly's lemma says this same deck determines every smaller proper subgraph count. The boundary is explicit: if $|F| = n$ the denominator $n - |F|$ vanishes, so the lemma reconstructs proper subgraph counts, not the full graph as a spanning object.

3.4 Reconstruction as a fiber problem

Definition 3.7 (Reconstructible graph)**.** An n-vertex graph G is reconstructible when $\delta_n^{-1}(\mathrm{Deck}(G)) = \{[G]\}$.

Definition 3.8 (Graph reconstruction conjecture)**.** The conjecture asserts that every finite simple graph with at least three vertices is reconstructible; equivalently, δ_n is injective for every $n \geq 3$.

Definition 3.9 (Section of the deck map)**.** A section is $s : \mathrm{im}(\delta_n) \to U_n$ with $\delta_n \circ s = \mathrm{id}_{\mathrm{im}(\delta_n)}$; it selects one representative class from each nonempty deck fiber.

Proposition 3.10 (Order asymmetry)**.** *For every section s, the composite $P_s = s\circ\delta_n$ is idempotent, $P_s^2 = P_s$, and equals the identity on U_n exactly when δ_n is injective.*

Proof. $P_s^2 = (s \circ \delta_n) \circ (s \circ \delta_n) = s \circ (\delta_n \circ s) \circ \delta_n = s \circ \mathrm{id} \circ \delta_n = P_s$. If $P_s = \mathrm{id}_{U_n}$ and $\delta_n([G]) = \delta_n([H])$, then $[G] = s(\delta_n([G])) = s(\delta_n([H])) = [H]$, so δ_n is injective. Conversely, if δ_n is injective, every fiber is a singleton, so every section returns the unique element of each fiber. □

3.5 Defective-calculus translation

The graph-theoretic data translate into Defective Calculus as follows. The translation respects decision (D1): the deck fiber carries the candidate class, while the retained kernel is the selection data that singles out a representative within it.

Defective-calculus object	Graph reconstruction object
Carrier $\mathcal{X}$	U_n, isomorphism classes of n-vertex graphs
Equivalence $\sim$	Graph isomorphism
θ-projection D_θ	Deck map δ_n
Observable $D_\theta([G])$	Vertex-deleted deck $\mathrm{Deck}(G)$
Fiber $F_\theta(D)$	Deck fiber $\delta_n^{-1}(D)$ (the candidate class)
Retained kernel $K_{\mathrm{ret}}([G])$	Selection datum singling out $[G]$ within $\delta_n^{-1}(\mathrm{Deck}(G))$
α-reconstruction I_α	Section / representative selection from a deck fiber
Polarity σ	Determined when fiber is a singleton; $\perp$ when not
Residual	Failure of a selected representative to be forced by the deck

Hence $D_\theta([G]) = \delta_n([G]) = \mathrm{Deck}(G)$, and the candidate (fiber) class is

$$F_\theta(\mathrm{Deck}(G)) = \delta_n^{-1}(\mathrm{Deck}(G)).$$

[Correction E5: the source set $K_{\mathrm{ret}}([G]) = \delta_n^{-1}(\mathrm{Deck}(G))$, which conflates the fiber with the kernel. Under Definition 2.5 with I_α = representative selection and $\sim$= isomorphism, $K_{\mathrm{ret}}([G])$ is the admissible selection data, while the candidate class is the fiber $F_\theta(\mathrm{Deck}(G))$. The reconstruction conjecture is a statement about the fiber *being a singleton, not about the kernel; this is the form decision (D1) fixes, and it removes the silent drift the source's final compression inherited.]*

The conjecture is therefore the fiber-singleton question $F_\theta(\mathrm{Deck}(G)) \stackrel{?}{=} \{[G]\}$. When the fiber is a singleton the graph is reconstructible and the retained kernel is trivial (no selection datum is needed). When the fiber is not known to be a singleton, DA records structural unresolved polarity, $\sigma(\mathrm{Deck}(G)) = \perp$ (decision D2): relative to the deck alone, no legal operation forces the representative.

3.6 What Kelly's lemma says in DA language

For every proper F with $|F| < n$, the count observable $O_F([G]) = N(F,G)$ factors through the θ-projection, $O_F = \tilde{O}_F \circ D_\theta$. Therefore all graphs in the same deck fiber share all proper subgraph counts:

$$D_\theta(G) = D_\theta(H) \implies O_F(G) = O_F(H) \quad \text{for all } |F| < n.$$

In defective-calculus language, proper subgraph counts do not belong to the unresolved part of the fiber—they have already survived the projection. The unresolved part, if any, is the residual representative choice, which the retained kernel would have to supply.

3.7 Failure surfaces

Code	Failure surface
G1	Treating Kelly's lemma as a proof of graph reconstruction.
G2	Confusing reconstructible parameters with reconstruction of the whole graph.
G3	Introducing labels, roots, or overlap identifications not present in the deck.
G4	Treating a section choice as canonical when the fiber is not known to be a singleton.
G5	Ignoring the boundary $\lvert F\rvert = n$, where Kelly's denominator vanishes.
G6	Using non-deck data to collapse $\sigma = \perp$.

3.8 Final compression

Kelly's lemma states that every proper subgraph count factors through the deck map:

$$N(F,G) = \frac{1}{n-|F|} \sum_{C \in \mathrm{Deck}(G)} N(F,C), \qquad |F| < n.$$

The reconstruction problem is the fiber problem $\delta_n^{-1}(\mathrm{Deck}(G)) \stackrel{?}{=} \{[G]\}$. In DA notation,

$$D_\theta = \delta_n, \qquad F_\theta(\mathrm{Deck}(G)) = \delta_n^{-1}(\mathrm{Deck}(G)), \qquad I_\alpha = \text{admissible representative selection},$$

with $K_{\mathrm{ret}}([G])$ the selection data inside that fiber. The worked example shows DA faithfully locates the fiber and kernel of graph reconstruction; it does not prove the conjecture.

References for this chapter. P. J. Kelly, *A congruence theorem for trees*, Pacific J. Math. **7** (1957), 961–968. S. M. Ulam, *A Collection of Mathematical Problems*, Interscience, 1960. J. A. Bondy and R. L. Hemminger, *Graph reconstruction—a survey*, J. Graph Theory **1** (1977), 227–268.

Chapter 4

Defective Analysis and the Numbers

Claim boundary. This chapter uses ordinary mathematical language: generator, action, quotient, kernel, group completion, localization, valuation, absolute value, metric completion, complexification, universal property, and presentation choice. The symbols $\theta, \alpha, \lambda, \rho$ are local notation for standard constructions and obstructions. By the clock-index doctrine (D4), θ and α here are clocks in the same sense as the spine—indices over the additive/successor and multiplicative/aggregation operations (§4.1)—not new structures. The terminal claim about $\mathbb{C}$ is a postulate with explicit proof obligations, not a completed theorem.

This chapter reconstructs the classical scalar ladder

$$0 \longrightarrow \{0,1\} \longrightarrow \mathbb{N} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{R} \longrightarrow \mathbb{C}$$

from a minimal generating presentation. The naturals arise from the forward successor action; the integers by group completion; the rationals by localization at nonzero integers; the reals by passing from presentation-dependent load to gauge-invariant size; the complexes by adjoining a polar unit so that exponentiation admits magnitude-preserving rotation rather than only real scaling.

4.1 The clock bridge

Before the construction, we discharge a notational obligation that the source documents left implicit. In the DA spine, θ is the access/derivation clock and α the reconstruction/aggregation clock. In the scalar ladder, θ is the successor action and α is multiplication. These are not two unrelated uses of the same letters; under the clock-index doctrine (D4) a clock is *an index over an allowed family of operations*, and:

- θ indexes *stepping/indexing access*—successor, the act of exposing the next element of a generated chain; and
- α indexes *aggregation/completion*—multiplication as the repeated-addition aggregate of copies, the same role the reconstruction clock plays upstream.

A superscript $+$ marks that only the forward operation is available; $\pm$ marks that the inverse has been adjoined. With this reading fixed once, the ladder may use θ, α exactly as the spine does:

$$\mathbb{N} = (\theta^+, \alpha^+), \quad \mathbb{Z} = (\theta^\pm, \alpha^+), \quad \mathbb{Q} = (\theta^\pm, \alpha^\pm), \quad \mathbb{R} = (\theta^\pm, \alpha^\pm, \lambda), \quad \mathbb{C} = (\theta^\pm, \alpha^\pm, \lambda, \rho).$$

Here λ is the completion obstruction selected by gauge-invariant, generator-visible size, and ρ is the polarization obstruction exposed by real exponentiation. *Guiding principle:* each new system is obtained by making a previously partial operation total, or by passing to a quotient/completion that removes presentation-dependence while preserving the intended scalar operations.

4.2 The zero stage and the minimal generating set

Let $S_{\min} = \{0, 1\}$ be the minimal generating set: 0 is neutral for addition, 1 generates the successor chain. The set $S_2 = \{0, 1, 2\}$ is *not* primitive; it is the first successor-closed extension because $2 = 1 + 1$. This distinction matters at the real-number stage: treating 2 as generated rather than primitive creates an apparent dyadic load, which will be shown to be presentation-dependent rather than scalar-invariant.

4.3 The natural numbers: forward additive generation

Definition 4.1 (Natural-number presentation)**.** The naturals are generated from $S_{\min}$ by the forward successor action $\theta(n) = n + 1$; equivalently $\mathbb{N} = \{0, 1, \theta(1), \theta^2(1), \dots\}$. The additive structure is the free commutative monoid on one generator, $(\mathbb{N}, +, 0) = \langle 1 \rangle_{\text{cmon}}$, and multiplication is repeated addition, $mn = \underbrace{n + \cdots + n}_{m}$. Thus $(\mathbb{N}, +, \cdot, 0, 1)$ is a commutative semiring.

The obstruction at this stage is that addition is not reversible: for $n > 0$, $n + x = 0$ has no solution in $\mathbb{N}$. The next system is forced if additive cancellation is to become internal.

4.4 The integers: group completion

Definition 4.2 (Grothendieck group completion)**.** Define $\mathbb{Z} = \mathbb{N} \times \mathbb{N}/\sim$ where $(a, b) \sim (c, d) \iff a+d = b+c$; the class of (a, b) is the formal difference $a-b$, with $n = [(n, 0)]$, $-n = [(0, n)]$. Addition and multiplication are $[(a, b)] + [(c, d)] = [(a + c, b + d)]$ and $[(a, b)] \cdot [(c, d)] = [(ac + bd, ad + bc)]$, well-defined on classes.

The result is the commutative ring $(\mathbb{Z}, +, \cdot, 0, 1)$; the positive successor action has become an additive group action, $\theta^+ \rightsquigarrow \theta^\pm$, so $\mathbb{Z} = (\theta^\pm, \alpha^+)$. The obstruction is that multiplication is not reversible for non-units: $2x = 1$ has no solution. The next system is forced if every nonzero scalar is to have a multiplicative inverse.

4.5 The rationals: localization by nonzero integers

Definition 4.3 (Localization of $\mathbb{Z}$)**.** Let $S = \mathbb{Z} \setminus \{0\}$. The rationals are $\mathbb{Q} = S^{-1}\mathbb{Z}$; equivalently $\mathbb{Q} = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\})/\sim$ with $(a, b) \sim (c, d) \iff ad = bc$, the class written a/b.

The result is the field $(\mathbb{Q}, +, \cdot, 0, 1)$; every nonzero element is invertible, so $\alpha^+ \rightsquigarrow \alpha^\pm$ and $\mathbb{Q} = (\theta^\pm, \alpha^\pm)$. The next obstruction is not a missing inverse—it is extracting a scalar notion of closeness from presentation-dependent generation data.

4.6 The completion obstruction: gauge variance of presentation load

The rationals carry the usual metric $d(x, y) = |x - y|$. A tempting argument observes that this metric does not control the complexity of generating a rational from a chosen presentation and then passes directly to the Cauchy completion. That passage is unforced: completing $\mathbb{Q}$ in $|\cdot|$ does not remove the disagreement between presentation loads, and the privileged status of $|\cdot|$ among possible metrics must be derived, not assumed. The corrected logic is:

1. presentation loads carry no gauge-invariant pointwise content (Lemma 4.6);
2. hence any notion of scalar closeness must be sourced in the operations, and the invariance axioms force such a function to be an absolute value (Lemma 4.10);
3. Ostrowski's classification together with a generator-visibility axiom selects the Archimedean absolute value uniquely up to uniform equivalence (Theorem 4.14);
4. $\mathbb{Q}$ is incomplete for that unique invariant uniformity (Proposition 4.16), and the completion is forced and unique (Theorem 4.18).

This is the obstruction λ, stated as a theorem chain rather than an analogy.

4.6.1 Presentations as gauges

Definition 4.4 (Presentation and load)**.** A presentation of $\mathbb{Q}$ is a finite subset $P \subset \mathbb{Q}$ with $\{0, 1\} \subseteq P$. The load $L_P(x)$ of $x \in \mathbb{Q}$ is the least $n \geq 0$ such that x is obtained from elements of P by n applications of $+, -, \times, \div$ (division by nonzero elements only). In particular $L_P(x) = 0$ iff $x \in P$.

Every $x \in \mathbb{Q}$ has finite load from every presentation, so all presentations generate the same field.

Definition 4.5 (Gauge moves)**.** An elementary gauge move adjoins a generated element, $P \mapsto P \cup \{g\}$ with $g \in \mathbb{Q}$, or deletes a redundant primitive, $P \mapsto P \setminus \{g\}$ with $g \notin \{0, 1\}$. Two presentations are gauge-equivalent if connected by elementary moves; since all presentations of $\mathbb{Q}$ generate $\mathbb{Q}$, all are gauge-equivalent. A quantity is gauge-invariant if it depends only on x, not on the presentation.

In this language $P_{\text{min}} = \{0, 1\}$ and $P_2 = \{0, 1, 2\}$ are two gauges and dyadic load is gauge-dependent; the next lemma makes the dependence total.

Lemma 4.6 (Loads are pure gauge)**.** *For every $x \in \mathbb{Q}$ and presentation P there is a gauge-equivalent P' with $L_{P'}(x) = 0$. Consequently $\inf_{P' \sim P} L_{P'}(x) = 0$ for every x, and any gauge-invariant $\ell : \mathbb{Q} \to [0, \infty)$ with $\ell \leq L_{P'}$ for all gauges P' is identically zero. There is no gauge-invariant "load of x."*

Proof. Take $P' = P \cup \{x\}$; the move is elementary because x is generated by P, and $L_{P'}(x) = 0$ by definition. The remaining claims follow at once. $\square$

Lemma 4.7 (The only invariant residue of load is properness)**.** *For every presentation P and every n, the sublevel set $\{x \in \mathbb{Q} : L_P(x) \le n\}$ is finite. Hence for every injective sequence (x_k) and every gauge P, $L_P(x_k) \to \infty$.*

Proof. Let $E_0 = P$ and let E_{k+1} be E_k together with all outputs of one operation applied to pairs from E_k. Then $|E_{k+1}| \le |E_k| + 4|E_k|^2 < \infty$ by induction, and $\{L_P \le n\} \subseteq E_n$. □

Lemma 4.7 is gauge-invariant but metrically empty: it declares every injective sequence heavy, so it cannot distinguish $b_n = 2^{-n}$ from 2^n.

Corollary 4.8 (Corrected reading of the dyadic example)**.** *Let $b_n = 2^{-n}$. Then $|b_n| \to 0$ while the dyadic load from $P_{\min}$ is n and from P_2 is 0. This is gauge variance of load, not a failure of the rational metric: no presentation load can serve as scalar size. If closeness is a property of scalars, it must be defined through the operations θ, α invariantly.*

4.6.2 Gauge-invariant size functions are absolute values

Loads obey $L_P(x \circ y) \le L_P(x) + L_P(y) + 1$ for $\circ \in \{+, \times\}$. *(Motivation, not a step in the chain:* exponentiating, any $s = b^{L_P}$ is submultiplicative and, after normalization, subadditive up to a constant; this only motivates the axioms below.*)* The gauge-invariant idealization of size is the following.

Definition 4.9 (Invariant size)**.** An invariant size on $\mathbb{Q}$ is $s : \mathbb{Q} \to [0, \infty)$ with

(S1) $s(x) = 0 \iff x = 0$ (non-degeneracy);

(S2) $s(xy) \le s(x)s(y)$ (compatibility with α);

(S3) $s(x + y) \le s(x) + s(y)$ (compatibility with θ);

(S4) $s(x^{-1}) = s(x)^{-1}$ for $x \ne 0$ ($\alpha^{\pm}$ symmetry).

Axiom (S4) is the metric expression of the fact that the ladder has already adjoined multiplicative inverses: a presentation may exhibit a scalar directly or via its inverse, and an invariant size must price the two reciprocally.

Lemma 4.10 (Multiplicativity)**.** *If s satisfies (S1), (S2), (S4), then $s(1) = 1$ and $s(xy) = s(x)s(y)$ for all x, y. Hence an invariant size is exactly an absolute value on $\mathbb{Q}$.*

Proof. By (S4) with $x = 1$, $s(1) = s(1)^{-1}$, so $s(1)^2 = 1$ and $s(1) = 1$. For $y \ne 0$, $s(x) = s\big((xy)y^{-1}\big) \le s(xy)s(y)^{-1}$ by (S2) and (S4), so $s(xy) \ge s(x)s(y)$; with (S2) this is equality. The case $y = 0$ is trivial by (S1). Finally $s(-1)^2 = s(1) = 1$ gives $s(-1) = 1$, so s is an absolute value in the usual sense. □

Theorem 4.11 (Ostrowski, 1916)**.** *Every absolute value on $\mathbb{Q}$ is either trivial, equivalent to the Archimedean $|\cdot|_\infty$, or equivalent to a p-adic $|\cdot|_p$ for exactly one prime p. Here $s \sim s'$ means $s' = s^c$ for some $c > 0$, equivalently that s, s' induce the same topology.*

This is cited, not reproved; it is the one classical input of this section. It says the gauge-invariant sizes on $\mathbb{Q}$ form a short exhaustive list. The dyadic phenomenon sits on that list: $L^{(2)}_{P_{\min}}(x) =$

$|\nu_2(x)| = \left|\log_2 |x|_2\right|$, so dyadic load is the logarithmic shadow of $|\cdot|_2$. The tension exhibited by $b_n = 2^{-n}$—null for $|\cdot|_\infty$ while $|b_n|_2 = 2^n \to \infty$—is, invariantly, the inequivalence of two absolute values. The selection problem is real and must be solved by an axiom.

4.6.3 Generator visibility selects the Archimedean branch

The whole ladder is generated by θ from 1. A size assigning bounded values to the successor orbit declares the generating action metrically invisible, which is incompatible with a program in which successor-depth is the primitive notion of magnitude.

Axiom 4.12 (Generator visibility)**.** $\sup_{n\in\mathbb{N}} s\big(\theta^n(0)\big) = \sup_{n\in\mathbb{N}} s(n) = \infty.$

Lemma 4.13 (Boundedness on $\mathbb{N}$ is the ultrametric dichotomy)**.** *For an absolute value s on $\mathbb{Q}$ the following are equivalent: (1) $s(n) \le 1$ for all $n \in \mathbb{N}$; (2) $s(x+y) \le \max\{s(x), s(y)\}$ for all x, y.*

Proof sketch. (2) $\Rightarrow$ (1) follows by induction from $s(1) = 1$. For (1) $\Rightarrow$ (2), expand by the binomial theorem and use multiplicativity:

$$s(x+y)^n \le \sum_{k=0}^{n} s\binom{n}{k} s(x)^k s(y)^{n-k} \le (n+1)\max\{s(x), s(y)\}^n,$$

then take n-th roots and let $n \to \infty$. □

Theorem 4.14 (Archimedean selection; uniqueness of the invariant uniformity)**.** *Let s be an invariant size on $\mathbb{Q}$ satisfying Axiom 4.12. Then $s = |\cdot|_\infty^c$ for some $c \in (0,1]$. All such s have the same null sequences and the same Cauchy sequences. Hence the gauge-invariant, generator-visible uniformity on $\mathbb{Q}$ exists and is unique.*

Proof. By Lemma 4.10, s is an absolute value. The trivial value and every $|\cdot|_p^c$ satisfy $s(n) \le 1$ on $\mathbb{N}$ (for the p-adics, $|n|_p \le 1$ since $n \in \mathbb{Z}$), so they violate Axiom 4.12. Ostrowski's theorem leaves $s = |\cdot|_\infty^c$ with $c > 0$. Subadditivity (S3) at $x = y = 1$ gives $2^c = s(2) \le 2$, so $c \le 1$. Since $s(x_n) \to 0$ iff $|x_n| \to 0$, and likewise for Cauchy tails, every admissible c induces the same uniform structure. □

Remark 4.15 (The dyadic shadow and the p-adic boundary)**.** Theorem 4.14 locates the p-adic completions precisely: they are what results from keeping (S1)–(S4) and dropping generator visibility. Each $|\cdot|_p$ is a gauge-invariant size whose completion $\mathbb{Q}_p$ is a complete valued field in which the successor tower is bounded; e.g. $2, 4, 8, \ldots$ is 2-adically null. The p-adic completions are not absorbed by the Archimedean completion; they are excluded from the classical ladder by Axiom 4.12.

4.6.4 Incompleteness of the invariant uniformity

Proposition 4.16 ($\mathbb{Q}$ is incomplete)**.** *There is a Cauchy sequence in $(\mathbb{Q}, |\cdot|_\infty)$ with no limit in $\mathbb{Q}$.*

Proof. Define $x_0 = 2$ and $x_{n+1} = \frac{x_n}{2} + \frac{1}{x_n}$, $e_n = x_n^2 - 2$. A direct computation gives $e_{n+1} = \left(\frac{e_n}{2x_n}\right)^2 \ge 0$. Moreover $e_n > 0$ for all n: if some x_n had $x_n^2 = 2$, then $\nu_2(x_n^2) = 2\nu_2(x_n)$ would be even while $\nu_2(2) = 1$ is odd. Hence $x_n^2 > 2$, so $x_n > 1$, and $x_n - x_{n+1} = \frac{e_n}{2x_n} > 0$. Thus the sequence is strictly decreasing with $1 < x_n \le 2$. From $x_n > 1$, $e_{n+1} \le \frac{e_n^2}{4}$ with $e_1 = \frac{1}{4}$, so $e_n \le 4^{1-2^{n-1}}$ for $n \ge 1$, and for $m < n$,

$$0 < x_m - x_n = \sum_{k=m}^{n-1} \frac{e_k}{2x_k} \le \frac{1}{2} \sum_{k \ge m} e_k \xrightarrow[m \to \infty]{} 0.$$

So the sequence is Cauchy in $\mathbb{Q}$. If it had a limit $q \in \mathbb{Q}$, then $q \ge 1 > 0$ and continuity of the rational operations away from 0 would give $q = \frac{q}{2} + \frac{1}{q}$, so $q^2 = 2$, contradicting the valuation-parity argument. □

Definition 4.17 (λ, revised)**.** The symbol λ denotes the obstruction and its repair: the unique gauge-invariant, generator-visible uniformity on $\mathbb{Q}$ is incomplete, and the repair is its metric completion. Presentation loads play no role in scalar identity; they are pure gauge.

4.7 The real numbers as the unique gauge-invariant completion

The completion is the standard Cauchy quotient. Let $\mathcal{C}(\mathbb{Q}) = \{(x_n) : x_n \in \mathbb{Q},\ (x_n) \text{ Cauchy}\}$ and $c_0(\mathbb{Q}) = \{(x_n) \in \mathcal{C}(\mathbb{Q}) : x_n \to 0\}$, and define $\mathbb{R} = \mathcal{C}(\mathbb{Q})/c_0(\mathbb{Q})$, with $(x_n) \sim (y_n) \iff x_n - y_n \to 0$. Operations descend: $[x] + [y] = [x + y]$, $[x][y] = [xy]$, $-[x] = [-x]$, and $\iota : \mathbb{Q} \hookrightarrow \mathbb{R}$, $q \mapsto [(q, q, \dots)]$. The quotient is a field: if $[x] \ne 0$ then x is not null, hence eventually bounded away from 0, and defining $y_n = 1$ for $n \le N$ and $1/x_n$ for $n > N$ gives $xy - 1 \in c_0(\mathbb{Q})$, so $[x][y] = [1]$. The order is $[x] > 0$ when some representative is eventually bounded below by a positive rational. The construction yields the complete Archimedean ordered field.

What is new is that the completion is not merely chosen; it is forced, and forced uniquely.

Theorem 4.18 (Necessity and uniqueness of $\mathbb{R}$)**.** *Let $K \supseteq \mathbb{Q}$ be any scalar system such that (i) the gauge-invariant, generator-visible size of Theorem 4.14 extends to an absolute value on K; (ii) every Cauchy sequence of K converges in K; (iii) K is minimal with (i) and (ii). Then K is isomorphic, by a unique isomorphism fixing $\mathbb{Q}$ and preserving the size, to $\mathbb{R} = \mathcal{C}(\mathbb{Q})/c_0(\mathbb{Q})$. Moreover $\mathbb{R}$ is, up to unique isomorphism, the only complete Archimedean ordered field, and it satisfies the universal property: every uniformly continuous map from $\mathbb{Q}$ into a complete metric space extends uniquely to $\mathbb{R}$.*

Proof sketch. The closure of $\mathbb{Q}$ in K is a completion of $(\mathbb{Q}, |\cdot|)$; by minimality it is all of K. Any two completions of a metric space are uniquely isometric over the space, and here the isometry is a field isomorphism because the operations are uniformly continuous on bounded sets and determined by their restrictions to the dense subfield $\mathbb{Q}$. Completeness plus the Archimedean property yields the complete Archimedean ordered field, whose uniqueness up to unique isomorphism is classical. The extension property is the universal property of metric completion. □

Remark 4.19 (The λ chain, displayed)**.** The transition $\mathbb{Q} \to \mathbb{R}$ is the composite of forced steps:

$$\underset{\text{Lem. 4.6}}{\text{loads are pure gauge}} \ \Rightarrow\ \underset{\text{Lem. 4.10}}{\text{size is an invariant absolute value}} \ \Rightarrow\ \underset{\text{Ostrowski}}{\{|\cdot|_\infty^c\} \cup \{|\cdot|_p^c\}}$$

$$\Rightarrow\ \underset{\text{Ax. 4.12}}{|\cdot|_\infty \text{ only}} \ \Rightarrow\ \underset{\text{Prop. 4.16}}{\mathbb{R}}.$$

In compact notation $\mathbb{R} = (\theta^\pm, \alpha^\pm, \lambda)$, but λ is no longer an interpretive label: it is the statement that gauge variance of presentation load, reduced to its invariant content, admits exactly one generator-visible uniformity, and that uniformity is incomplete over $\mathbb{Q}$.

4.8 Exponentiation as a coupled additive–multiplicative action

After $\mathbb{R}$ is obtained, both operations are reversible and completion is available. Yet a remaining obstruction appears when additive parameters are coupled to multiplicative action by exponentiation. Real exponentiation $\exp : \mathbb{R} \to \mathbb{R}_{>0}$, $\exp(t) = e^t$, is a one-parameter multiplicative subgroup with $e^{s+t} = e^s e^t$, but it is line-bound: $e^t > 0$ for all t. It scales; it does not rotate. If the scalar system is required to support a magnitude-preserving polar one-parameter action, the real line is insufficient. This is the obstruction ρ.

4.9 The complex numbers: polar complexification

Definition 4.20 (Complexification of $\mathbb{R}$)**.** Adjoin i with $i^2 = -1$; equivalently $\mathbb{C} = \mathbb{R}[x]/(x^2+1)$. Every element is $z = a + bi$ with $a, b \in \mathbb{R}$; as a real vector space $\mathbb{C} \cong \mathbb{R}^2$, with $(a, b)(c, d) = (ac - bd,\ ad + bc)$.

The polar unit repairs the exponentiation obstruction: $e^{it} = \cos t + i \sin t$ with $|e^{it}| = 1$, giving the magnitude-preserving rotation $z \mapsto e^{it}z$. Thus $\mathbb{R} \to \mathbb{C}$ is not only the algebraic solution of $x^2 + 1 = 0$—that equation is the minimal-polynomial symptom—but the structural repair of polar complexification: real scaling extended to include unit-magnitude rotation. In compact notation $\mathbb{C} = (\theta^\pm, \alpha^\pm, \lambda, \rho)$. The standard fact behind the terminal role of $\mathbb{C}$ is algebraic closure, used only inside the claim boundary below.

4.10 The full scalar ladder

System	Structure	Construction added	Public reading
0	none	none	null starting stage
$S_{\min}$	$0, 1$	minimal generating set	identity and unit generator
$\mathbb{N}$	θ^+, α^+	successor closure	forward generation; semiring
$\mathbb{Z}$	$\theta^\pm, \alpha^+$	group completion	additive inverses; ring
$\mathbb{Q}$	$\theta^\pm, \alpha^\pm$	localization	multiplicative inverses; field
$\mathbb{R}$	$\theta^\pm, \alpha^\pm, \lambda$	selected completion	generator-visible completion
$\mathbb{C}$	$\theta^\pm, \alpha^\pm, \lambda, \rho$	complexification	polar unit; algebraically closed

4.11 Universal properties

$\mathbb{N}$**:** $(\mathbb{N}, +, 0)$ is the free commutative monoid on one generator: for every commutative monoid M and $m \in M$ there is a unique homomorphism $f : \mathbb{N} \to M$ with $f(1) = m$. $\mathbb{Z}$**:** the group completion of $\mathbb{N}$: every monoid homomorphism $f : \mathbb{N} \to G$ into an abelian group factors uniquely through $\mathbb{Z}$. $\mathbb{Q}$**:** the localization of $\mathbb{Z}$ at nonzero integers: every ring homomorphism $f : \mathbb{Z} \to R$ sending nonzero integers to units factors uniquely through $\mathbb{Q}$. $\mathbb{R}$**:** the metric completion of $\mathbb{Q}$ under the unique gauge-invariant, generator-visible uniformity of Theorem 4.14; every uniformly continuous map from $\mathbb{Q}$ into a complete metric space extends uniquely to $\mathbb{R}$. $\mathbb{C}$**:** the quadratic extension $\mathbb{R}[x]/(x^2+1)$ and the algebraic closure of $\mathbb{R}$; it supplies the polar unitary group $U(1) = \{z \in \mathbb{C} : |z| = 1\} = \{e^{it} : t \in \mathbb{R}\}$.

4.12 Terminal scalar postulate

Postulate 4.21 (Terminal classical scalar postulate)**.** Subject to: (1) the system is generated from $S_{\min} = \{0, 1\}$; (2) addition and multiplication are commutative scalar operations with identities; (3) additive inverses are adjoined by group completion; (4) multiplicative inverses for nonzero scalars by localization; (5) scalar size is gauge-invariant in the sense of (S1)–(S4), generator-visible in the sense of Axiom 4.12, and Cauchy convergence has endpoints; (6) exponentiation is extended to include magnitude-preserving polar rotation; (7) equivalence is up to isomorphism or harmless change of presentation—then $\mathbb{C}$ is the terminal classical scalar layer forced by these requirements.

This postulate does not claim no other algebraic systems exist; it does not exclude finite fields, p-adic fields, nonstandard reals, surreal numbers, quaternions, Clifford algebras, matrix or operator algebras, schemes, sheaves, or higher categories. Those alter the hypotheses, change the notion of scalar, abandon commutativity, use a different absolute value, add dimension rather than scalar closure, or move to a different category. The claim is only that the ladder $\mathbb{N} \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{R} \to \mathbb{C}$ is terminal for the stated classical scalar program.

4.13 Open questions and proof obligations

1. **Category of presentations.** Define the category whose objects are scalar presentations and whose morphisms preserve $0, 1$, the operations, completion data, and polar structure.
2. **Load invariance beyond $\mathbb{Q}$.** Characterize the analogous invariant load-like functions for broader scalar presentations.
3. **p-adic branches.** Classify the branch obtained when generator visibility is weakened or replaced.
4. **Equivalence of real completions.** Relate the Cauchy construction, Dedekind cuts, and nested intervals inside the presentation-size argument.
5. **Polarization criterion.** Specify the exact universal property that forces $\mathbb{C}$ from $\mathbb{R}$: algebraic closure, polar exponentiation, two-dimensional rotation, or a combined condition.
6. **Boundary systems.** Classify precisely why p-adics, quaternions, nonstandard reals, surreals, and operator algebras fall outside the postulate or require a modified one.
7. **Terminality theorem.** Convert the postulate into a theorem by proving that any scalar system satisfying the hypotheses factors through $\mathbb{C}$ uniquely.

4.14 Final compression

$$\mathbb{C} = S_{\min} + \theta^{\pm} + \alpha^{\pm} + \lambda + \rho.$$

In public language: $\mathbb{C}$ = minimal generating set + additive inverses + multiplicative inverses + gauge-invariant, generator-visible completion + polar complexification. The number systems are a sequence of forced repairs to partial scalar structure: $\mathbb{N}$ generates; $\mathbb{Z}$ reverses addition; $\mathbb{Q}$ reverses multiplication; $\mathbb{R}$ completes the selected invariant uniformity; $\mathbb{C}$ polarizes exponentiation.

References for this chapter. A. Ostrowski, *Über einige Lösungen der Funktionalgleichung* $\varphi(x)\varphi(y) = \varphi(xy)$, Acta Math. **41** (1916), 271–284. S. Lang, *Algebra*, rev. 3rd ed., Springer, 2002. W. Rudin, *Principles of Mathematical Analysis*, 3rd ed., McGraw-Hill, 1976.

Chapter 5

A Formal Exploration of Geometry

Claim boundary. This chapter is an exploratory reading of standard geometric structures—point, incidence, affine, metric, topological, smooth, tangent, curvature, geodesic, fiber, gauge, complex—in the vocabulary of typed carrier, projector, quotient, and residual fiber. It proves no new geometric theorem and re-derives nothing classical. The formulas displayed are the standard definitions; the DA gloss is interpretive. Each section keeps its declared "known unknown" (an open modeling choice) and "unknown unknown" (a question the framework cannot yet pose precisely).

The organizing thesis is that geometry, read through DA, is the admissible reconstruction of relational defect over a typed carrier rather than a primitive ambient space. A geometric object is specified relative to a carrier, coefficient structure, witness data, equivalence, projectors, quotient layer, retained kernel, and a residual comparator:

$$\mathcal{G} = (X,\ A,\ W,\ E,\ \Pi,\ Q,\ K,\ R_{\mathrm{cmp}}), \qquad D_\theta X = (Q, K), \qquad I_\alpha(Q, K) \sim_E X.$$

The quotient Q is the exposed geometric layer; the retained kernel K is the hidden fiber needed to reconstruct what the quotient discarded. The diagnostic for a real (non-gauge) fiber is

$$I_\alpha(Q, 0) \not\sim_E X \implies K \text{ is geometrically real.}$$

5.1 Terminal claim

Geometry is here treated not as a pre-existing ambient space but as the stabilized reconstruction of relations among distinguishable witnesses after quotienting and residual retention. A mature DA geometry states its type, carrier, coefficient structure, projectors, quotient, residual fiber, and admissibility condition, $\mathcal{G} = (X, A, W, E, \Pi, Q, K, R_{\mathrm{cmp}})$, with $D_\theta X = (Q, K)$ and $I_\alpha(Q, K) \sim_E X$.

5.2 Geometry from the algebraic ladder

The scalar ladder of Chapter 4 repairs operations; the geometric ladder reads those repairs as relational structure. The same ascent induces a parallel geometry ladder:

$$0 \to \mathbb{N} \to \mathbb{Z} \to \mathbb{Q} \to \mathbb{R} \to \mathbb{C},$$

$$0 \to \mathrm{Pt} \to \mathrm{Ray}_{\mathbb{N}} \to \mathrm{Line}_{\mathbb{Z}} \to \mathrm{Aff}_{\mathbb{Q}} \to \mathrm{Geom}_{\mathbb{R}} \to \mathrm{PhaseGeom}_{\mathbb{C}}.$$

Reading the stages: 0 is no exposed distinction; Pt a first localized distinction; $\mathrm{Ray}_{\mathbb{N}}$ forward discrete extension; $\mathrm{Line}_{\mathbb{Z}}$ oriented reversible extension; $\mathrm{Aff}_{\mathbb{Q}}$ ratio/intersection geometry; $\mathrm{Geom}_{\mathbb{R}}$ continuous metric and smooth geometry; $\mathrm{PhaseGeom}_{\mathbb{C}}$ rotation, phase, and holomorphic closure.

5.3 The point

A point is read as a stabilized equivalence class of local witnesses rather than a primitive atom: $p = [w]_E$ with $w \in W$, and $I_\alpha(D_\theta p) \sim_E p$, written $0 \rightsquigarrow \mathrm{Pt}$. It is the first geometric repair of the zero class: pure zero exposes no localization, and the point repairs indistinguishability by exposing location.

Known unknown. Whether DA should treat points as primitive witnesses, quotient classes of witnesses, or terminal reconstruction residues. *Unknown unknown.* Whether there are regimes where "point" is always a bad chart and must be replaced by region, fiber, event, or process.

5.4 Incidence geometry

Given a coefficient structure A, the affine A-module is $V_A = A^n$; a line through p in direction v is the orbit $L(p, v) = \{p + tv : t \in A\}$, and incidence is the relation $p \in L$. Incidence asks which objects meet—not distance, angle, smoothness, or curvature. For $A = \mathbb{N}$ incidence is forward-only; for $\mathbb{Z}$ reversible; for $\mathbb{Q}$ it admits rational interpolation and slope; for $\mathbb{R}$ it becomes continuum-stable; for $\mathbb{C}$ it carries phase and rotation.

Known unknown. Whether incidence is the first geometric relation or whether adjacency/preorder comes earlier. *Unknown unknown.* Whether some DA geometries admit incidence without points, using only morphisms or overlaps.

5.5 Affine geometry

Affine geometry appears when translation is admissible; it preserves collinearity and ratios but not length. In DA terms it is geometry modulo origin choice: the origin is a gauge choice, the difference vector $v = q - p$ is retained, and absolute position is quotient-relative. Thus affine geometry is already residual-fibered, with Q the relative-position data and K the discarded origin fiber:

$$\mathrm{Aff}(A^n) = A^n \rtimes \mathrm{GL}_n(A), \qquad x \mapsto Mx + b,\ M \in \mathrm{GL}_n(A),\ b \in A^n.$$

Known unknown. Whether the affine origin is a killed gauge fiber or a retained reconstruction kernel. *Unknown unknown.* Whether DA prefers affine to vector geometry because absolute origin is usually non-observable.

5.6 Metric geometry

A metric is read as a residual comparator: distance is the cost of reconstructing one witness from another under a selected chart, not primitive separation. The standard axioms hold,

$$d : X \times X \to \mathbb{R}_{\geq 0}, \quad d(x,y) = 0 \iff x \sim_E y, \quad d(x,y) = d(y,x), \quad d(x,z) \leq d(x,y) + d(y,z),$$

with $d(x,y)$ the reconstruction cost of x from y under the chosen chart. A Cauchy path whose limit escapes X records a metric defect, repaired by completion $\bar{X} = \mathrm{Comp}(X,d)$.

[Correction E12: the source wrote "metric geometry requires $\mathbb{R}$." That is too strong: metric and pseudometric structures are defined over any ordered value carrier, and non-Archimedean (p-adic), rational-valued, and lattice-valued metrics all exist. What completion actually needs is a complete *value carrier for the chosen size. The corrected statement: limit-stable comparison requires a complete value carrier; for the generator-visible Archimedean size of Chapter 4 that carrier is $\mathbb{R}$, but the requirement is completeness of the value object, not $\mathbb{R}$ specifically.]*

Known unknown. Whether every DA residual comparator should be a metric, pseudometric, asymmetric cost, divergence, or preorder. *Unknown unknown.* Whether physical distance is a special case of reconstruction cost or whether reconstruction cost is only an epistemic proxy.

5.7 Topological geometry

A topology is read as the grammar of admissible neighborhoods; an open set is a perturbation-stable reconstruction region. The axioms are standard: $\varnothing, X \in \tau$; arbitrary unions and finite intersections of members of τ lie in τ. Continuity, $f^{-1}(V) \in \tau_X$ for $V \in \tau_Y$, is read as: f does not convert small admissible input defects into inadmissible output defects.

Known unknown. Whether topology should be derived from residual comparison or admitted as a primitive stability grammar. *Unknown unknown.* Whether DA supports non-classical topologies where neighborhoods are determined by projector sectors rather than subsets.

5.8 Smooth geometry

A smooth manifold is read as a multi-chart reconstruction object: smoothness means the derivative clock survives chart transition,

$$\varphi_i : U_i \to \mathbb{R}^n, \qquad \varphi_j \circ \varphi_i^{-1} \text{ smooth where defined}, \qquad M = (|M|, \tau, \{\varphi_i\}, E, \Pi, K),$$

with $D_\theta(\varphi_j \circ \varphi_i^{-1})$ existing and remaining admissible. A singularity appears where no admissible chart absorbs the residual: $p \in \mathrm{Sing}(M)$ iff for every admissible chart, local reconstruction near p fails.

Known unknown. Whether DA singularities should be classified by failed chart, projector, quotient, or retained kernel. *Unknown unknown.* Whether some physical singularities are artifacts of forcing smooth charts where DA would use a different admissible category.

5.9 Tangent structure

The tangent space at p is read as the first-order defect sector of curves through p—the local grammar of possible departures. For $\gamma : (-\varepsilon, \varepsilon) \to M$ with $\gamma(0) = p$, the velocity $v = \dot{\gamma}(0) \in T_pM$, and T_pM is the space of admissible first-order departures; the tangent bundle $TM = \bigsqcup_{p \in M} T_pM$ is then the geometry of possible local defects over the base.

Known unknown. Whether tangent vectors should be reconstructed from curves, derivations, germs, infinitesimal probes, or sectorized witness changes. *Unknown unknown.* Whether DA admits tangent structure in non-smooth settings without forcing classical differentiability.

5.10 Curvature

Curvature is read as the obstruction to path-independent reconstruction—the residual of transporting local reconstruction around a loop:

$$R_\nabla(X, Y)Z = \nabla_X \nabla_Y Z - \nabla_Y \nabla_X Z - \nabla_{[X,Y]}Z, \qquad R_\nabla = [\nabla_X, \nabla_Y] - \nabla_{[X,Y]}.$$

Flatness ($R_\nabla = 0$) is path-independent loop reconstruction; nonzero curvature is a loop residual. This is the geometric face of the DA clock-order principle (D3): derivative and reconstruction clocks need not commute, $D_\theta I_\alpha \neq I_\alpha D_\theta$.

Known unknown. Whether DA curvature is best identified with connection curvature, quotient curvature, holonomy residual, or projector noncommutativity. *Unknown unknown.* Whether all curvature-like effects unify as failure of operation-order invariance.

5.11 Geodesics

A geodesic is read as a locally self-reconstructing path: $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$, i.e. local continuation introduces no avoidable acceleration defect. In metric geometry geodesics locally extremize length; in DA geometry they minimize reconstruction correction under the chosen connection.

Known unknown. Whether DA geodesics should be defined variationally, connection-theoretically, or by minimum residual accumulation. *Unknown unknown.* Whether some systems carry several inequivalent geodesic notions depending on projector sector.

5.12 Fiber geometry

A fiber bundle separates exposed base structure from retained local freedom: with $\pi : E \to B$ and fiber $F_b = \pi^{-1}(b)$, the base is the quotient ($Q = B$) and the fiber is the retained kernel ($K = F$). A gauge choice is a section $s : B \to E$ with $\pi \circ s = \mathrm{id}_B$ (a local trivialization). Gauge freedom is read not as meaningless redundancy but as retained-kernel structure invisible to the quotient yet necessary for reconstruction.

Known unknown. Whether all DA retained kernels should be modeled as fibers. *Unknown unknown.* Whether some retained kernels are not bundle-like and require stacks, sheaves, higher categories, or nonlocal memory objects.

5.13 Gauge curvature

Given a gauge connection A, its curvature $F_A = dA + A \wedge A$ measures the defect of local fiber alignment under transport: $F_A = 0$ means the gauge field is locally pure reconstruction choice; $F_A \neq 0$ means the retained fiber has observable residual. The reading offered is that physics enters geometry when retained reconstruction fibers become dynamically constrained.

Known unknown. Whether DA gauge curvature is fundamental or a special case of projector-sector holonomy. *Unknown unknown.* Whether DA can classify physical interactions by the kind of retained kernel their quotient observables require.

5.14 Complex geometry

Complex geometry is read as the stage where the real line is insufficient to reconstruct phase: the complex coordinate retains real magnitude and adds phase closure, $\mathbb{C} = \mathbb{R}[i]/(i^2 + 1)$, $z = x + iy$, with rotation $z \mapsto e^{i\theta} z$. A holomorphic function ($\bar{\partial} f = 0$) respects the phase structure rather than treating x, y as independent real axes once the phase fiber is admitted.

Known unknown. Whether complex structure is phase closure, orientation closure, or the minimal algebraic repair of real curvature defects. *Unknown unknown.* Whether DA derives quantum phase from complex geometry or whether complex geometry is one representation of a deeper phase fiber.

5.15 Singularities, typed

A singularity is read as a failed admissible reconstruction site, not an infinite value, and—importantly—must be typed by the sector that fails rather than collapsed into one untyped failure:

$$p \in \mathrm{Sing}(X) \iff \text{no available chart makes } p \text{ reconstructible with finite typed residual,}$$

with subtypes $\mathrm{Sing}_{\mathrm{chart}}$ (coordinate failure), $\mathrm{Sing}_{\mathrm{metric}}$ (distance/completion failure), $\mathrm{Sing}_{\mathrm{smooth}}$ (derivative failure), $\mathrm{Sing}_{\mathrm{fiber}}$ (retained-kernel failure), and $\mathrm{Sing}_{\mathrm{quotient}}$ (over-compression failure).

Known unknown. Whether singularities should be repaired by completion, resolution, blow-up, category change, or retained-kernel restoration. *Unknown unknown.* Whether physical singularities are real ontic breakdowns or inadmissible quotient compression.

5.16 COTG as geometric quotient

For a geometric object X, the Whole Defect System W_X contains more than visible geometry—charts, fibers, transitions, residuals, clocks, projector sectors. The COTG layer (Chapter 6) is the exposed curvature-parity quotient, not identical with the WDS; the lost structure is the kernel of the WDS-to-COTG projection:

$$\mathrm{COTG}(X) = \Pi(W_X)/E, \qquad \mathrm{COTG}(X) \neq W_X, \qquad K_X = \ker\bigl(W_X \to \mathrm{COTG}(X)\bigr),$$

with $I_\alpha(\mathrm{COTG}(X), K_X) \sim_E X$. Mature DA geometry is therefore always exposed quotient plus retained fiber plus residual comparison.

5.17 The full DA geometry ladder

$$
\begin{gathered}
0 \to \mathrm{Pt} \to \mathrm{Inc} \to \mathrm{Aff} \to \mathrm{Met} \to \mathrm{Top} \to \mathrm{Diff} \\
\to \mathrm{Curv} \to \mathrm{Fib} \to \mathrm{Gauge} \to \mathrm{Phase}.
\end{gathered}
$$

Reading the rungs: Pt localized distinction; Inc incidence; Aff relative position; Met reconstruction cost; Top perturbation-stable neighborhoods; Diff stable derivative clock across charts; Curv noncommuting reconstruction around loops; Fib retained kernel over a quotient base; Gauge fiber-choice symmetry with observable curvature residual; Phase complex/polar closure.

5.18 Final compression

$$
\begin{gathered}
\mathcal{G} = \mathrm{AdmRec}_{\mathrm{DA}}(X; A, \theta, \alpha, \Pi, E, Q, K, R_{\mathrm{cmp}}), \qquad D_\theta X = (Q, K), \qquad I_\alpha(Q, K) \sim_E X, \\
I_\alpha(Q, 0) \not\sim_E X \implies K \text{ is geometrically real.}
\end{gathered}
$$

Read in one line: geometry is what arithmetic becomes when defects between elements must be reconstructed relationally—the residual fiber of failed flat reconstruction.

Chapter 6

Charted Oscillatory Transport Geometry

Claim boundary. This chapter defines Charted Oscillatory Transport Geometry (COTG) as a realization layer derived from Defective Analysis through Whole Defect Systems (WDS). The WDS path is primary; the parity quotient $K_2 \oplus K_{\mathrm{odd}}$ is a secondary construction route, not the whole definition. The two-sector probe is an example, not the definition. The mathematics used—finite-dimensional complex vector spaces, unitary phase multipliers, tensor products, Gram matrices, and inner-product expansions—is standard. No external application is claimed.

COTG is the charted oscillatory realization layer of DA. A WDS supplies an upstream carrier, a defect extractor, defect sectors, a gluing carrier, and a quotient/gluing projection. Under a parity-stable quotient, non-parity defect sectors may be collapsed while a binary gluing kernel is preserved. The analytic COTG layer then studies local oscillatory charts, unitary phase transport, tensor aggregation, gluing residuals, projection, and admissibility. The central separation is between latent phase transport, which preserves magnitude, and projection, which can expose relative phase through interference.

6.1 Position in the DA hierarchy

$$\mathrm{DA} \longrightarrow \mathrm{WDS} \longrightarrow D_P(\mathcal{N}) = \Delta\mathcal{N} \longrightarrow \text{optional parity quotient} \longrightarrow \text{COTG realization.}$$

COTG is not the whole upstream system. The WDS supplies the upstream defect object; COTG supplies the chart-level analytic realization for charted oscillatory transport, projection, and residual compatibility.

6.2 Whole Defect Systems in DA language

Definition 6.1 (Whole Defect System)**.** A Whole Defect System is a tuple $\mathcal{N} = (X, \Delta, \{K_p\}_{p\in P}, G, q)$ where X is the carrier of states before quotient; Δ is the defect-extraction operator; K_p is a defect sector indexed by an internal label p; G is the gluing or comparison carrier;

and q is the quotient or gluing projection identifying defect directions invisible at the declared resolution.

The WDS is a DA object because it declares a carrier, a defect extractor, a quotient structure, and a comparison carrier; its validity depends on the category in which these data are interpreted.

Definition 6.2 (Primal derivative)**.** The Primal derivative of a WDS is the defect extractor $D_P(\mathcal{N}) = \Delta\mathcal{N}$. It exposes the defect object and the sector data on which quotienting or gluing acts.

Definition 6.3 (Primal integral)**.** The Primal integral is reconstruction from a quotient or gluing skeleton, generally only modulo a killed kernel: $I_P(\text{quotient data}) = \mathcal{N} \bmod K_{\text{killed}}$. It does not reconstruct a unique pre-quotient WDS unless the relevant retained kernel is supplied.

6.3 The parity quotient as a secondary construction

Some WDS instances admit a sector decomposition $\Delta\mathcal{N} = \bigoplus_{p\in P} K_p$. When one sector is distinguished as a binary/parity sector K_2, set $K_{\text{odd}} = \bigoplus_{p\in P,\, p>2} K_p$, where $p > 2$ is internal grammar notation for the non-parity sector labels.

Definition 6.4 (Odd-sector equivalence)**.** States $x, y \in X$ are odd-sector equivalent, $x \sim_{\text{odd}} y$, when $\Delta(x) - \Delta(y) \in K_{\text{odd}}$. The quotient is $Q_2(\mathcal{N}) = X/\!\sim_{\text{odd}}$.

Definition 6.5 (Parity projection)**.** The induced parity defect projection is $\Delta_2([x]) = \text{proj}_{K_2}(\Delta x)$, well-defined only when odd-sector equivalence does not change the K_2 component:

$$x \sim_{\text{odd}} y \implies \text{proj}_{K_2}(\Delta x) = \text{proj}_{K_2}(\Delta y).$$

This is the parity no-leakage condition.

Proposition 6.6 (Parity quotient route)**.** *If $\mathcal{N}$ is a WDS with $\Delta\mathcal{N} = K_2 \oplus K_{\text{odd}}$ and the quotient kills exactly K_{odd} while preserving K_2, then the parity-stable quotient residue is $Q_2(\mathcal{N}) \simeq \mathcal{N}/K_{\text{odd}}$ in the declared quotient category.*

Remark 6.7. The parity quotient is a construction route, not the entire COTG definition. COTG is the charted oscillatory realization of the quotient or of another DA/WDS object satisfying the same admissibility requirements.

6.4 Finite COTG object

Definition 6.8 (Finite COTG realization)**.** A finite COTG realization is a tuple $\mathcal{C} = (M, \mathcal{A}, E, \{X_i\}_{i\in I}, \{q_i\}_{i\in I}, \{g_{ij}\}_{(i,j)\in E}, T, \Pi, R)$ with M an index space / base chart domain; $\mathcal{A} = \{(U_i, \varphi_i)\}$ a finite chart family (atlas) on M; E a directed overlap edge set between chart domains; X_i a local complex state space (fiber) over U_i; $q_i(t)$ a local oscillatory section on U_i; g_{ij} a transition-like map on an overlap edge (i,j); T a global tensor object built from local fibers; Π a projection or measurement map into observable space Y; and R residual data measuring gluing, compatibility, or projection failure.

[Correction E8 (notation): the source used the symbol A for the chart family, again for a transport operator in the example, and A_k for tensor amplitudes. Here the atlas is $\mathcal{A}$, the transport operator in the example is A, and tensor amplitudes are A_k, so the three never collide. The object is admitted by declared structure—chart overlaps, local fibers, transition maps, tensor object, projection, residual law—not by name.]

6.5 Local oscillatory charts

Definition 6.9 (Local oscillator)**.** For each chart U_i, let X_i be a finite-dimensional complex vector space with basis $\{e_{i1}, \dots, e_{iN_i}\}$. A local oscillatory section is the finite mode sum

$$q_i(t) = \sum_{k=1}^{N_i} a_{ik} \exp\bigl(i(\omega_{ik}t + \varphi_{ik})\bigr) e_{ik}.$$

The coefficients a_{ik} determine latent magnitude; the phase terms determine relative transport and projected interference.

6.6 Phase transport and latent magnitude

Definition 6.10 (Off-axis phase-time displacement)**.** An off-axis coordinate α_i determines a displacement $\Delta t_i = \lambda_i \alpha_i$, and substitution gives

$$q_i(t + \lambda_i\alpha_i) = \sum_k a_{ik} \exp\bigl(i(\omega_{ik}t + \varphi_{ik})\bigr) \exp(i\omega_{ik}\lambda_i\alpha_i) e_{ik}.$$

Lemma 6.11 (Local magnitude invariance)**.** *Local phase-time displacement preserves each local mode magnitude.*

Proof. Each coefficient is multiplied by $\exp(i\vartheta)$ with $\vartheta \in \mathbb{R}$; since $|\exp(i\vartheta)| = 1$, the magnitude is unchanged. □

6.7 Global tensor state

The finite tensor space is $T = \bigotimes_{i\in I} X_i$, and the shifted global state is $Q(t;\alpha) = \bigotimes_{i\in I} q_i(t + \lambda_i\alpha_i)$. After expansion into a global tensor basis $\{E_k\}_{k\in K}$,

$$Q(t;\alpha) = \sum_{k\in K} A_k \exp\bigl(i(\Omega_k t + \Phi_k + \Theta_k(\alpha))\bigr) E_k,$$

where $A_k = \prod_i a_{ik_i}$, $\Omega_k = \sum_i \omega_{ik_i}$, $\Phi_k = \sum_i \varphi_{ik_i}$, and $\Theta_k(\alpha) = \sum_i \omega_{ik_i}\lambda_i\alpha_i$.

Definition 6.12 (Latent magnitude object)**.** The finite latent magnitude object is $M_Q = (|A_k|)_{k\in K}$.

Remark 6.13 (Global latent-magnitude invariance)**.** For every mode k, $\bigl|A_k \exp(i\psi_k)\bigr| = |A_k|$, so M_Q is independent of t and α under phase-only transport.

[Correction E8 (status): the source labeled the line above "Theorem 1." It is an immediate restatement of the local magnitude lemma at the global tensor level ($|e^{i\psi}| = 1$ applied modewise), with no further deductive content, so it is recorded as a remark. The word "theorem" is reserved here for the projected-norm expansion below, which carries the actual interference content.]

6.8 Gluing residuals

Definition 6.14 (Transition-like map)**.** On an overlap edge (i, j), a transition-like map has the form $\tau_{ij} = \lambda_j\alpha_j - \lambda_i\alpha_i$, $g_{ij} = F_{ij} \circ S_{\tau_{ij}}$, where $S_{\tau_{ij}}$ is the phase-time shift component and F_{ij} the fiber/state component.

Definition 6.15 (Finite phase residual on frequency-matched pairs)**.** For a mode pair (ik, jl) the full relative phase is

$$\Delta\psi_{ij}^{kl}(t, \alpha) = (\omega_{ik} - \omega_{jl})\, t + (\varphi_{ik} - \varphi_{jl}) + (\omega_{ik}\lambda_i\alpha_i - \omega_{jl}\lambda_j\alpha_j) \pmod{2\pi}.$$

On a *frequency-matched* pair, $\omega_{ik} = \omega_{jl}$, the t-term drops and the residual is the time-independent quantity

$$R_{ij}^{kl}(\alpha) = \omega_{ik}\lambda_i\alpha_i + \varphi_{ik} - \omega_{jl}\lambda_j\alpha_j - \varphi_{jl} \pmod{2\pi}.$$

Full gluing also requires amplitude, frequency, and basis compatibility, recorded as $D_{ij}^{kl} = (A_{ij}^{kl}, W_{ij}^{kl}, B_{ij}^{kl}, R_{ij}^{kl})$.

[Correction E7: the source wrote $R_{ij}^{kl}(\alpha)$ as if always time-independent. In general the relative phase carries a term $(\omega_{ik} - \omega_{jl})t$, which vanishes only when the pair is frequency-matched. The definition above states the residual on frequency-matched pairs and exposes the $(\Omega)t$ term otherwise, so the "residual" is not silently assumed t-independent.]

6.9 Projection and observable interference

Let $\Pi : T \to Y$ be a linear map into a finite-dimensional inner-product space, and adopt the convention that the Hermitian inner product $\langle\cdot, \cdot\rangle$ is conjugate-linear in its first argument and linear in its second. The projected observable is

$$y(t; \alpha) = \Pi Q(t; \alpha) = \sum_k A_k \exp(i\psi_k(t, \alpha))\, v_k, \qquad v_k = \Pi(E_k).$$

Theorem 6.16 (Projected interference formula)**.** *With the inner-product convention above, the projected squared norm decomposes as*

$$\begin{aligned} \|y(t; \alpha)\|^2 &= \sum_{k,l} \overline{A_k}\, A_l\, \exp\bigl(i(\psi_l - \psi_k)\bigr)\langle v_k, v_l\rangle \\ &= \sum_k |A_k|^2\|v_k\|^2 + \sum_{k\neq l} \overline{A_k} A_l\, e^{i(\psi_l - \psi_k)}\langle v_k, v_l\rangle. \end{aligned}$$

If the amplitudes are taken in the polar gauge $a_{ik} \in \mathbb{R}_{\geq 0}$ (hence $A_k \in \mathbb{R}_{\geq 0}$), then $\overline{A_k}A_l = A_kA_l$ and the diagonal is $\sum_k A_k^2\|v_k\|^2$.

Proof. Expand $\langle y, y\rangle$ with $y = \sum_k c_kv_k$, $c_k = A_ke^{i\psi_k}$. By conjugate-linearity in the first slot, $\langle y, y\rangle = \sum_{k,l} \overline{c_k}c_l\langle v_k, v_l\rangle$, and $\overline{c_k}c_l = \overline{A_k}A_le^{i(\psi_l - \psi_k)}$. The diagonal $k = l$ gives $\overline{A_k}A_k\langle v_k, v_k\rangle = |A_k|^2\|v_k\|^2$; the off-diagonal terms carry relative phases and projected Gram entries $\langle v_k, v_l\rangle$. $\square$

[Correction E6: the source wrote $\sum_{k,l} A_k A_l e^{i(\psi_k-\psi_l)}\langle v_k, v_l\rangle$, dropping the conjugation on A_k and on the phase. For complex amplitudes this is not the squared norm. The corrected form conjugates the first factor ($\overline{A_k}A_l$, phase $\psi_l-\psi_k$); the source's expression is recovered exactly in the declared polar gauge $A_k \geq 0$, which is the case the rest of the document implicitly used. The diagonal $\sum_k |A_k|^2\|v_k\|^2$ is unchanged either way.]

Corollary 6.17 (Steering criterion)**.** *The projected observable is phase-steerable by α exactly when the off-diagonal interference functional is nonconstant in α.*

6.10 Arbitrary-dimensional extension

The arbitrary-dimensional version replaces finite vector spaces by complex topological vector spaces H_i, finite sums by convergent spectral series or direct integrals, finite tensor products by completed tensor products or replacement objects, and finite Gram matrices by kernels or quadratic forms. A spectral representation may be discrete, $q_i(t) = \sum_{k\in K_i} a_i(k)\exp(i(\omega_i(k)t + \varphi_i(k)))e_{i,k}$, or continuous, $q_i(t) = \int_{\Sigma_i} a_i(\sigma)\exp(i(\omega_i(\sigma)t + \varphi_i(\sigma)))e_{i,\sigma}\, d\mu_i(\sigma)$. Phase transport must be a well-defined unitary multiplier on the chosen state class or dense domain, $(U_{i,\alpha_i}f)(\sigma) = \exp(i\omega_i(\sigma)\lambda_i\alpha_i)f(\sigma)$.

6.11 Intrinsic admissibility

A candidate realization is admitted by reconstruction of structure, not by name, and must specify: a *base* whose chart intersections reproduce the overlap structure or gluing category; *fibers* preserving mode, spectral, or generalized basis data; *transitions* whose reconstructed diagrams commute on common domains; *transport* represented by unitary multipliers for phase-only motion; *projection* defined on the transported state family and preserving Gram/kernel data; *residual* compatibility of the state class with gluing, transport, invariants, and projection; a *limit* specification of state, magnitude, and observable convergence for finite-to-infinite passage; and, if a parity quotient is invoked, *quotient* preservation of the parity sector under odd-sector collapse.

6.12 Operational example: two-sector alpha phase

This example illustrates the operational use of the definition; it is not the definition. Let A be a fixed transport operator on a hidden but probeable finite system; the carrier is fixed by the θ-clock, and the α-clock shifts the relative phase between two legal probe sectors. With $z = z_+ + z_-$ a decomposition into two legal sectors, set $z(\phi) = z_+ + e^{i\phi}z_-$, $\phi \in [-\pi, \pi]$. The scalar response is $R_\phi(k) = z(\phi)^* A^k z(\phi)$, and expanding,

$$R_\phi(k) = z_+^* A^k z_+ + z_-^* A^k z_- + e^{i\phi} z_+^* A^k z_- + e^{-i\phi} z_-^* A^k z_+.$$

A global phase on all of z cancels, but a relative phase between sectors can change the projected scalar response; phase sensitivity is detected when $\partial R_\phi(k)/\partial\phi \neq 0$ for some k. The legal-probe rule is $z \in \mathrm{Sector}_{\alpha,\theta}(P)$, with P the declared measurement-port space: no hidden support is directly accessed, only inferred from scalar-response constraints.

6.13 Failure surfaces

Code	Failure surface
C1	Undefined chart overlaps or gluing category.
C2	Mode mismatch: amplitude, frequency, basis, or spectral compatibility missing.
C3	Nonunitary phase transport: claimed phase-only motion changes latent magnitude.
C4	Operator-domain failure: projection, transition, or residual operator undefined.
C5	Divergent expansion in the declared topology.
C6	Unspecified tensor completion or replacement object.
C7	Projection-kernel failure: the Gram/kernel expression is undefined.
C8	False steering: projected interference constant despite a steering claim.
C9	Uncontrolled finite-to-infinite passage.
C10	Quotient ambiguity: parity quotient invoked without preserving the parity sector.
C11	COTG/WDS conflation: COTG treated as the full WDS rather than a realization layer.

6.14 Known unknowns and proof obligations

1. Fix the ambient category for each WDS instance.
2. Specify the algebraic status of sector decompositions such as $K_2 \oplus K_{\text{odd}}$.
3. Prove no-leakage for any invoked quotient.
4. Declare tensor completions and operator domains in infinite-dimensional examples.
5. Characterize when operational phase-sector probes are minimal and admissible.

6.15 Final compression

COTG is the DA realization layer for charted oscillatory transport:

$$\text{local charts} + \text{unitary phase transport} + \text{gluing residuals} + \text{tensor aggregation} + \text{projection} + \text{admissibility}.$$

The WDS path is $\mathcal{N} \to D_P(\mathcal{N}) = \Delta\mathcal{N} \to$ sector data $\to$ optional parity quotient $\to$ COTG realization, and the central analytic separation is that latent phase transport preserves magnitude while projection can expose relative phase.

Chapter 7

Global Definitions, Operator Cores, and Repository Discipline

Claim boundary. This chapter states proposed internal definitions and document-control rules in ordinary mathematical language: set, class, map, quotient, fiber, kernel, section, residual, reconstruction, equivalence, admissibility, grammar, and extraction record. The symbols are local notation. No claim is made that the framework is complete, correct, novel, universal, or applicable outside the declared formal setting. The purpose is to prevent drift between the DA spine and the surrounding material.

This chapter records the objects adjacent to the spine: WDS as a standalone primitive, the Primal/Scalar Calculus hierarchy, Reconstruction Calculus as the operator core, the original COTG skeleton, generator seeds, the clock-coordinate doctrine, the clock-order residual, retained-kernel admissibility, and thread/glossary discipline. It is a source-index layer for keeping drafts typed, reviewable, and non-leaking, not a new theorem.

7.1 Purpose and status

Definition 7.1 (Document admissibility)**.** A DA document is document-admissible when every term it uses is either defined in that document, imported from a named upstream document, or marked provisional with an explicit proof obligation.

The spine is DA $\to$ DC $\to$ worked examples $\to$ WDS $\to$ COTG; this chapter records the surrounding definitions seeds $\to$ WDS $\to$ PC $\to$ SC, with Recon $\subseteq$ DC $\subseteq$ DA read as a grammar containment (Remark 7.10).

7.2 Whole Defect Systems as a standalone primitive

Definition 7.2 (Whole Defect System, full form)**.** A Whole Defect System is a tuple $\mathcal{N} = (\mathcal{A}, X, \Delta, \{K_p\}_{p\in P}, G, q, \mathrm{Adm}_{\mathcal{N}})$ where $\mathcal{A}$ is the declared ambient category; X the carrier before quotient; Δ the defect-extraction operator; K_p a defect sector indexed by $p \in P$; G the gluing/comparison carrier; q the quotient/gluing projection; and $\mathrm{Adm}_{\mathcal{N}}$ the admissibility predicate.

The shorter tuple $\mathcal{N} = (X, \Delta, \{K_p\}_{p\in P}, G, q)$ is admissible notation only when $\mathcal{A}$ and $\mathrm{Adm}_{\mathcal{N}}$ are fixed by context.

Definition 7.3 (Defect object and sector decomposition)**.** The defect object is the extracted data $\Delta\mathcal{N}$ (an obstruction or distinguishability datum, not necessarily numerical). A WDS admits a sector decomposition when $\Delta\mathcal{N} \simeq \bigoplus_{p\in P} K_p$; distinguishing a binary sector, $\Delta\mathcal{N} \simeq K_2 \oplus K_{\mathrm{odd}}$ with $K_{\mathrm{odd}} = \bigoplus_{p>2} K_p$, where $p > 2$ is internal sector grammar unless an external ordered index has been separately constructed.

7.3 Primal Calculus and Scalar Calculus

Definition 7.4 (Primal Calculus)**.** Primal Calculus is the partial operator calculus paired with WDS and COTG data: $\mathrm{PC} = (\mathrm{Sys}, \mathrm{WDS}, \mathrm{COTG}, \mathrm{Gen}_P, D_P, D_0, I_P, I_0, K_{\mathrm{ret}}, \mathrm{Adm}_P)$. The operators are partial; a candidate system is not primal-admissible merely because it has been named.

Definition 7.5 (Primal derivative and integral)**.** For a WDS presentation $\mathcal{N}$, $D_P(\mathcal{N}) = \Delta\mathcal{N}$. For a candidate system S supplied with a WDS presentation and comparison data, $D_P(S) = (C_S, K_S) \in \mathrm{Gen}_P$, with C_S the admitted COTG object and K_S a typed residual generator. The complete Primal integral is partial reconstruction $I_P(C, K) \to S/{\sim_S}$; the flat integral $I_0(C) \to S/(\sim_S \vee K_{\mathrm{hid}})$ omits the retained generator. Thus I_0 reconstructs only modulo the hidden sector, while I_P may reconstruct up to system equivalence when K is admissible.

Definition 7.6 (Scalar Calculus)**.** Scalar Calculus is the scale-indexed test sector. For a scale index s, $\mathrm{SC}_s = (X_s, \sim_s, K_s, \Delta_s, P_s, \mathrm{COTG}_s, K_{s,\mathrm{ret}}, I_s, \mathrm{Adm}_s)$, with intended grammar $D_s(X) = \mathrm{WDS}_s(X)$, $\mathrm{COTG}_s(X) = K_s/K_{s,\mathrm{hid}}$, $I_s(X) = (\mathrm{COTG}_s(X), K_{s,\mathrm{ret}}(X))$, and scalar admissibility $\mathrm{Recon}_s(\mathrm{COTG}_s(X), K_{s,\mathrm{ret}}(X)) \sim_s X$. It is the finite or scale-indexed layer in which quotient, retained-kernel, and no-leakage claims are made concrete, not a replacement for Primal Calculus.

7.4 Reconstruction Calculus as the operator core

Definition 7.7 (Reconstruction Calculus instance)**.** A Reconstruction Calculus instance is a tuple $\mathcal{R} = (X, \sim, Y, K, D, I, \Pi, \mathrm{Res}, \mathrm{Adm})$ where $D : X \to Y$ is a defective observable, $I : Y \times K \to X/{\sim}$ a reconstruction operation, Π a visible quotient/comparison map, Res a declared residual carrier or map, and Adm an admissibility predicate.

Definition 7.8 (Flat and complete reconstruction)**.** If $0 \in K$ is a declared null retained datum, flat reconstruction is $I^0 D(X) = I(DX, 0)$; complete reconstruction is any $I(DX, k)$ with $k \in K_{\mathrm{ret}}(X)$. The central reconstruction test is $I(DX, k) \sim X$.

Definition 7.9 (Reconstruction equivalence)**.** The base reconstruction relation is

$$X \; R_0 \; X' \iff DX = DX' \text{ and } \exists\, k, k' \in K \left[\mathrm{Adm}(X, k) \wedge \mathrm{Adm}(X', k') \wedge I(DX, k) \sim I(DX', k')\right].$$

Reconstruction equivalence $\equiv_{\mathcal{R}}$ is the *transitive closure* of R_0: $X \equiv_{\mathcal{R}} X'$ iff there is a finite chain $X = Z_0 \; R_0 \; Z_1 \; R_0 \cdots R_0 \; Z_n = X'$. Equivalently, in the stricter realization that fixes a section s or demands equality in a declared quotient object Q (i.e. $\Pi_Q I(DX, k) = \Pi_Q I(DX', k')$), the base relation is already transitive and the two notions coincide.

[Correction E9: as the source wrote it (Definition 12), reconstruction equivalence used an unquantified existential witness on each side. Reflexivity and symmetry hold, but transitivity can fail: a witness pair joining X to X' and another joining X' to X'' need not compose into a witness joining X to X'', since the two reconstructions of X' may differ. Taking the transitive closure (or the section/quotient realization, which is transitive outright) restores an honest equivalence relation. This matters wherever $\equiv_{\mathcal{R}}$ is used to form a quotient.]

Remark 7.10 (Containment of grammar)**.** Reconstruction Calculus is the operator core of Defective Calculus; Defective Calculus adds clock labels, polarity, static/operational regime separation, and no-leakage discipline, and DA adds clock-signature framing: Recon $\subseteq$ DC $\subseteq$ DA. This is a containment of grammars (each later layer declares strictly more structure), not an assertion about subfields of standard mathematics, and not a theorem.

[Correction E11: the source labeled the containment "Proposition 1." Nothing is deduced; it records how three declared vocabularies nest. It is stated here as a remark so that "proposition" continues to mark a claim with a proof.]

7.5 Original COTG skeleton

Definition 7.11 (Original COTG skeleton)**.** The original COTG skeleton is the minimal parity-stable cyclic gluing object $\mathrm{COTG}(\mathcal{N}) = (A, B, C; E; K_2; \mathrm{glue}_2)$, where A, B, C are comparison sectors, E an equality carrier, K_2 the surviving parity kernel, and glue_2 the binary gluing law. If $\mathcal{N}$ is a WDS with $\Delta\mathcal{N} \simeq K_2 \oplus K_{\mathrm{odd}}$ and quotienting by K_{odd} preserves K_2, the parity-stable route is $\mathrm{COTG}(\mathcal{N}) \simeq \mathcal{N}/K_{\mathrm{odd}}$ in the declared quotient category.

The skeleton is provisional notation, usable only when its equality carrier, parity kernel, and gluing law are supplied. The matured ordering is WDS first, defect extraction second, parity quotient third, COTG realization fourth.

7.6 Generator seeds and seed gauge

Definition 7.12 (Seed gauge)**.** A seed gauge is a declared primitive generator presentation. The two basic gauges are $G_{01} = \{0, 1\}$ and $G_{012} = \{0, 1, 2\}$: the first treats 2 as generated by $1 + 1$, the second treats 2 as present at the seed level. These are different presentations even when they generate isomorphic terminal arithmetic objects under later equivalence.

Postulate 7.13 (Seed no-leakage)**.** A reconstruction may use seed-generated data only through operations legal in the declared seed gauge. If 2 is generated, it may be used once the generating operation is supplied; if 2 is primitive, that primitivity must be declared as input data. Until the gauge is declared, claims about primitivity, generation, minimality, and retained kernel remain underdetermined.

7.7 Clock coordinates

Definition 7.14 (Clock coordinate)**.** A clock coordinate is an allowed operation axis—an index over a family of admissible operations. It is not a parameter of ordinary time. It records how a system may be accessed, transformed, reconstructed, glued, or compared.

By the clock-index doctrine (D4), the word *coordinate* in "clock coordinate" abbreviates *operation-index axis* and is the clock sense of index; it is distinct from a geometric coordinate (the dimension sense) and from ordinary time (a third index). The θ-coordinate is the derivation/access axis, $D_\theta : X \to Y$, answering "what is legally exposed?"; the α-coordinate is the reconstruction axis, $I_\alpha : Y \times K \to X/\sim$, answering "what is legally completed from the exposed data and retained kernel?" The clock signature is $\mathrm{Clk}(P) \subseteq \{\theta, \alpha\}$, with the diagnostic local $= \theta \veebar \alpha$, global $= \theta \wedge \alpha$, and failure $=$ wrong assignment, exactly as in Chapter 1.

7.8 Clock-order residual

By decision (D3) the noncommutativity of the clocks is fundamental, so the residual that records it must be well-typed. The primary clock-order residual keeps both terms in the observable carrier Y.

Definition 7.15 (Clock-order residual, primary form)**.** For $k \in K$, the clock-order residual in the observable carrier is

$$\Delta_{\theta,\alpha}(X; k) \;=\; D_\theta\, I_\alpha(D_\theta X, k) \;-\; D_\theta X \;\in\; R_Y,$$

where both $D_\theta I_\alpha(D_\theta X, k)$ and $D_\theta X$ live in Y (then a declared residual carrier R_Y). It records the obstruction produced when the exposed data are reconstructed with retained datum k and re-exposed. The flat order residual is the same with $k = 0$: $\Delta^0_{\theta,\alpha}(X) = D_\theta\, I_\alpha(D_\theta X, 0) - D_\theta X$.

Definition 7.16 (Commutator shorthand, secondary)**.** If a declared comparison functor sends both a reconstruction in $X/\sim$ and an exposure in Y into one common carrier R, one may write the shorthand $E_{\theta,\alpha}(X) = \mathrm{Res}_R(D_\theta I_\alpha(\,\cdot\,), I_\alpha D_\theta(\,\cdot\,))$, and, when R is additive, $E_{\theta,\alpha}(X) = D_\theta I_\alpha(X) - I_\alpha D_\theta(X)$. This is admissible only after both terms and the retained datum are typed in the same carrier.

[Correction E10: the source's Definition 23 wrote the clock-order residual directly as a comparison of $D_\theta I_\alpha$ and $I_\alpha D_\theta$. As written this is ill-typed: I_α takes two arguments (y, k) and returns an element of $X/\sim$, while $D_\theta I_\alpha(\,\cdot\,)$ returns an element of Y, so the two composites live in different spaces and cannot be compared until both are mapped into a common carrier and *a retained datum is fixed. The repair makes $\Delta_{\theta,\alpha}(X; k)$ (both terms in Y) the primary object—matching Correction E3 in Chapter 2—and demotes the $D_\theta I_\alpha$ vs $I_\alpha D_\theta$ commutator to a shorthand contingent on a declared comparison functor. The same repair applies to the compound-time residual in Chapter 8 and to the clock-order remark in Chapter 1.]*

Remark 7.17. A complete integral may be reversible by construction when the retained kernel is sufficient. The noncommutativity claim therefore belongs first to the clock-order residual of incomplete, flat, or incompatible reconstructions; a complete reconstruction still has to prove that the relevant diagram closes (decision D3).

7.9 Retained kernel, admissibility, reconstruction equivalence

Definition 7.18 (Retained-kernel class)**.** For a Defective Calculus instance with $D_\theta : X \to Y$ and $I_\alpha : Y \times K \to X/\sim$, $K_{\mathrm{ret}}(X) = \{k \in K : I_\alpha(D_\theta X, k) \sim X \text{ and } \mathrm{Adm}(X, k)\}$.

Definition 7.19 (Admissible reconstruction; no-leakage)**.** A reconstruction is admissible when (1) the observable was produced by a declared D_θ; (2) it uses only declared I_α operations; (3) retained data lie in K_{ret}; (4) no-leakage holds; (5) the result equals the target up to $\sim$, or the residual is recorded. No-leakage holds when every quantity used is either contained in the stated data or generated by a declared legal operation; hidden data may be retained but not smuggled into visible certainty.

Postulate 7.20 (Retained-kernel discipline)**.** A retained kernel is an admissible reconstruction class, not a rescue channel. If K is expanded to make a reconstruction succeed, the expansion must be typed, must preserve no-leakage, and must reject at least one nearby inadmissible reconstruction.

7.10 Thread and glossary discipline

Definition 7.21 (Thread extraction record)**.** A thread extraction record is a tuple $T = (H, C, N, S, O, F, A, D)$: source header H, core-claims list C, new-object list N, structural changes S, omissions O, failure surfaces F, action items A, and extraction digest D.

Definition 7.22 (Glossary entry)**.** A glossary entry is a tuple $G_t = (t, d, s, r, u, a)$: term t, definition d, stability status s, replacement/deprecated wording r, upstream source u, ambiguity risk a.

Postulate 7.23 (Glossary no-drift rule)**.** A term may not silently change meaning across documents. If a term changes, the later document must mark the change as a correction, restriction, extension, or replacement.

7.11 Failure surfaces

Code	Failure surface
A1	WDS used as a word without declaring its tuple data.
A2	COTG treated as the whole WDS instead of a quotient/realization layer.
A3	Primal Calculus stated as a universal inverse rather than a partial factorization.
A4	Scalar Calculus used without scale index, quotient, retained kernel, equivalence.
A5	Reconstruction succeeds only because K was expanded into a rescue channel.
A6	Seed gauge not declared, so generated and primitive data are conflated.
A7	θ, α used as decoration rather than typed operation coordinates.
A8	Clock-order residual written as a commutator before both composites are typed.
A9	Thread notes imported as definitions without stability, dependencies, failures.
A10	Glossary drift changes a term without declaring a correction or replacement.

7.12 Known unknowns and proof obligations

1. Specify the categories in which WDS tuples, quotient objects, sector decompositions, and reconstruction maps live.
2. Determine when θ, α can be derived from an object rather than assigned.

3. Characterize which sector decompositions are genuine direct sums, which are filtrations, and which are only presentation labels.
4. State the universal property, if any, of the original COTG skeleton $(A, B, C; E; K_2; \mathrm{glue}_2)$.
5. Separate the seed-gauge problem (primitive presentation) from the arithmetic-ladder problem (generated closure).
6. Define the minimal residual carrier for clock-order residuals without forcing all examples into an additive setting.
7. Give finite neighboring examples where retained-kernel admissibility succeeds in one case and fails in another.
8. Create a glossary versioning rule strict enough to prevent drift but not so strict that exploratory notes become impossible.

7.13 Unknown-unknown controls

Unknown unknowns cannot be listed directly, so the discipline is a set of control surfaces: every future document should state its carrier, equivalence, legal operations, quotient, retained kernel, residual carrier, admissibility rule, source status, and glossary changes. A missing item is an open defect, not tacit permission.

7.14 Final compression

$$\mathrm{DA}_{\mathrm{global}} = (\text{seed gauge}, \mathrm{WDS}, \mathrm{Recon}, \mathrm{DC}, \mathrm{PC}, \mathrm{SC}, \mathrm{COTG}, K_{\mathrm{ret}}, \mathrm{Adm}, \text{thread discipline}).$$

The operational spine is $G \to \mathcal{N} \xrightarrow{\Delta} \Delta\mathcal{N} \to \text{sector data} \to \mathrm{COTG} \to (\mathrm{COTG}, K_{\mathrm{ret}}) \xrightarrow{I} X/\!\sim$. The no-leakage spine is: visible quotient + admissible retained kernel $\neq$ arbitrary hidden rescue data. The document-control spine is: thread extraction $\to$ glossary entry $\to$ typed draft $\to$ reviewable failure surface.

Chapter 8

A Physics-Unification Schema

Claim boundary. This chapter states a proposed internal unification *schema* for DA. It is not an experimentally confirmed theory of nature, a completed derivation of quantum mechanics, a completed derivation of general relativity, or a replacement for existing quantum-gravity programs. It is a typed reconstruction argument: *if* physics is represented as admissible repair under legal clocked operators, *then* quantum mechanics, general relativity, and thermodynamics appear as local chart-realizations of one parent action–entropy selection structure. Every result below is internal to that representation. The chapter is for confirmation, criticism, and formal review; its open obligations and falsifiers are preserved in full.

DA describes a problem by assigning operation clocks, stating what a legal projection exposes, and specifying which retained data may reconstruct without leakage. This chapter formulates a unification schema in which the parent carrier is not a space of particles, metrics, strings, or events, but the class of admissible reconstructions. The θ-clock exposes curvature, projection loss, boundary data, or observable defect; the α-clock supplies admissible histories, gluing, and reconstruction. Time is treated as a compound operator built from the ordered interaction of exposure and repair; entropy as the logarithmic measure of admissible-repair multiplicity in a terminal class; action as an α-clock equivalence class. A local physical theory is then a symmetry-class realization of a parent action–entropy selector.

8.1 Purpose and current status

The reconciliation of gravity and quantum theory remains unresolved. Recent work still treats the quantum nature of gravity and the relation between general relativity and quantum mechanics as an open problem, with proposed experiments and theoretical routes under active investigation [11,10,9].[1] String theory offers one major route via extended microscopic degrees of freedom and compactification; in recent reviews, four-dimensional unification arises if the extra dimensions are compactified suitably [12]. DA begins instead from: what must be retained, reconstructed, and

[1]Reference [10] (Aziz and Howl, *Classical theories of gravity produce entanglement*, Nature **646**, 813–817, 2025) is cited here as evidence that the seam is live, not as a settled result: several groups (Gundhi–Diósi-type re-calculations and Marletto–Oppenheim–Vedral–Wilson) argued in late 2025–2026 that the reported entanglement is an artifact of discarded transition amplitudes and that classical gravity does not mediate entanglement. The dispute itself *supports* the “open problem” framing used here rather than resolving it in either direction.

quotiented for a visible physical state to be well-formed? The thesis is structural, not triumphal: physics can be *represented* as terminal admissible repair under compounded clocks, and under that representation QM, GR, and thermodynamics are local symmetry charts of one reconstruction-selection structure.

8.2 Primitive DA instance

Definition 8.1 (DA instance and clock coordinates)**.** A DA instance is $\mathcal{D} = (X, \sim, Y, K, D_\theta, I_\alpha, \Pi, \sigma, \mathrm{Adm})$ as in Chapter 2, with $D_\theta : X \to Y$ the θ-access observable and $I_\alpha : Y \times K \to X/\sim$ the α-reconstruction. By the clock-index doctrine (D4) these are operation clocks (operation-index axes), not parameters of ordinary time. The retained-kernel class is $K_{\mathrm{ret}}(X) = \{k \in K : I_\alpha(D_\theta X, k) \sim X \text{ and } \mathrm{Adm}(X, k)\}$: data invisible under projection but legal for reconstruction, not an arbitrary rescue channel. No-leakage is as before.

8.3 The unifying carrier: admissible reconstructions

The parent object is not a state space in the ordinary physical sense; it is the space of admissible reconstructions, $X_{\mathrm{Phys}} := \mathrm{AdmRec}$. A physical state is not a bare representative but the terminal equivalence class of representatives indistinguishable after all legal clocks/operators have acted.

Definition 8.2 (Legal operator system)**.** Let the legal clocked and chart operators be $\theta, \alpha, q, g, \tau, \ldots$ (here q denotes quantum-chart operations, g geometric-chart, τ thermodynamic/coarse-graining, the ellipsis other admitted charts), composed whenever the compositions are meaningful. A word O is legal only if every intermediate carrier is typed and admissible.

[Correction E16: the source called this generated structure a "monoid" $\Omega = \langle \theta, \alpha, q, g, \tau, \ldots \rangle$. Composition here is partial*—only defined when the intermediate carriers type-match and are admissible—so the structure is not a monoid (whose composition is total). It is a* typed partial monoid*, equivalently the collection of morphisms of a* category *$\mathcal{Q}$ whose objects are admissible typed carriers and whose morphisms are legal operators, with composition defined exactly when codomain and domain agree. "$O \in \Omega$" below abbreviates "O a legal endomorphism-chain in $\mathcal{Q}$ from the state's carrier to an observable carrier."]*

Definition 8.3 (Terminal physical equivalence)**.** For admissible reconstructions x, x', $x \sim_\Omega x' \iff O(x) = O(x')$ for every legal compounded observable O. The terminal physical state is $\Psi_{\mathrm{Phys}} = [x]_\Omega \in X_{\mathrm{Phys}}/\sim_\Omega$: a physical state is a terminal equivalence class, not a private representative.

Proposition 8.4 (Defect-quotient criterion)**.** *A defect δ quotients out under the full operator system iff $O(x) = O(x + \delta)$ for every legal compounded O; equivalently $\delta \in K_\Omega \iff x \sim_\Omega x + \delta$. If this fails, the defect is visible to at least one legal compounded operator and must be retained as physical structure.*

Proof. This is the definition of $\sim_\Omega$ applied to $(x, x + \delta)$: if all legal observables agree, the defect is invisible in the terminal quotient; if some legal observable differs, the pair lies in distinct terminal classes and the defect cannot be discarded without leakage. $\square$

8.4 Theta curvature, alpha histories, compound time

DA separates exposure from reconstruction: θ is the curvature/projection clock, α the history/reconstruction clock. The θ-clock exposes boundary, curvature, projection loss, differentiation, measurement, or coarse-grained defect; the α-clock supplies admissible histories.

Definition 8.5 (Compound time)**.** Time is not identified with θ or α alone. The primary compound-time residual is taken in the observable carrier, for a retained datum k,

$$\Delta^{\mathrm{T}}_{\theta,\alpha}(X;k) = D_\theta\, I_\alpha(D_\theta X, k) - D_\theta X \in R_Y,$$

both terms in Y. When a declared comparison functor sends a reconstruction in $X/\sim$ and an exposure in Y into a common carrier R, one may use the secondary order-pair shorthand $\mathbf{T} = (D_\theta I_\alpha, I_\alpha D_\theta)$ with $E_{\theta,\alpha}(X) = \mathrm{Res}_R(D_\theta I_\alpha(X), I_\alpha D_\theta(X))$, additive form $D_\theta I_\alpha(X) - I_\alpha D_\theta(X)$, only after both terms are typed in the same carrier.

[Correction E10 (carryover): the source's compound-time definition wrote $\mathbf{T} = (D_\theta I_\alpha, I_\alpha D_\theta)$ and compared $D_\theta I_\alpha(X)$ with $I_\alpha D_\theta(X)$ directly. As in Chapter 7, these composites land in different spaces (Y versus $X/\sim$) and need a common carrier and a retained datum before comparison; the well-typed primary object is $\Delta^{\mathrm{T}}_{\theta,\alpha}(X;k)$ above. Ordinary time is then a local chart of the compound interaction between visible access and admissible history, not a primitive.]

8.5 Clock-order entropy

Definition 8.6 (Admissible repair set and repair measure)**.** For a terminal state $[x]_\Omega$, the admissible repair set is $\mathcal{R}([x]_\Omega) = \{r \in X_{\mathrm{Phys}} : r \sim_\Omega x \text{ and } \mathrm{Adm}(r)\}$. For a clock pair (θ, α) with residual typed into a common carrier and a retained repair class $K^{\mathrm{ret}}_{\theta\alpha}([x])$, let $\mu_{\theta\alpha}$ be a declared measure or counting rule on $K^{\mathrm{ret}}_{\theta\alpha}([x])/\sim$ (its existence and normalization are a proof obligation, item 4 of §8.16).

Definition 8.7 (Clock-order entropy)**.** The entropy associated with the compound clock is

$$S_{\theta\alpha}([x]) = \log \mu_{\theta\alpha}\big(K^{\mathrm{ret}}_{\theta\alpha}([x])/\sim\big) :$$

the logarithmic measure of admissible-repair multiplicity remaining after visible identification.

Proposition 8.8 (Vanishing-residual criterion)**.** *If $D_\theta I_\alpha$ and $I_\alpha D_\theta$ are typed into a common residual carrier and the retained repair class is a singleton modulo $\sim$, then the clock-order residual is trivial and $S_{\theta\alpha}([x]) = \log 1 = 0$. If the orders fail to commute, the residual cannot be removed by visible data alone; the legal completions are the elements of $K^{\mathrm{ret}}_{\theta\alpha}$ satisfying admissibility, and quotienting by $\sim$ identifies repairs differing only by invisible representative choice, leaving multiplicity $\mu_{\theta\alpha}(K^{\mathrm{ret}}_{\theta\alpha}/\sim)$ whose logarithm is the entropy. This matches the usual pattern: entropy counts compatible hidden alternatives relative to a declared visible state.*

[Correction E13: the source stated this as "Theorem 1" but the displayed relation is the definition *of $S_{\theta\alpha}$, and its "proof" restates that definition and notes the singleton case. The deductive content is exactly the vanishing-residual criterion (singleton repair class $\Leftrightarrow$ zero entropy), recorded as a proposition; the entropy expression is a definition. The measure $\mu_{\theta\alpha}$, undeclared in the source, is now an explicit datum with a standing obligation to construct it without circularity.]*

In plain terms: projection exposes a defect; history repairs it; order matters; multiplicity remains; multiplicity is entropy.

8.6 Parent entropy and local symmetry realizations

Definition 8.9 (Parent and chart entropy)**.** With μ_Ω a measure on admissible repair classes, the parent entropy of a terminal state is $S_\Omega([x]) = \log \mu_\Omega(\mathcal{R}([x]_\Omega)/\sim_\Omega)$. For a local chart c with symmetry class G_c, the chart entropy is $S_c([x]) = \log \mu_c(\mathcal{R}([x]_\Omega)/G_c)$.

Proposition 8.10 (Entropy realization principle)**.** *Different entropy formulas arise when the same parent repair multiplicity is quotiented by different chart symmetries. Shannon, Boltzmann, von Neumann, and geometric entropy are therefore to be treated as chart-dependent realizations until a stricter equivalence theorem is supplied.*

Chart	Symmetry / quotient	Local entropy reading
Information	partitions / codes	uncertainty over admissible messages compatible with visible data
Thermodynamics	macrostate coarse-graining	multiplicity of micro-repairs consistent with macro-observables
Quantum mechanics	unitary / polarity structure	mixed-state or branch multiplicity after admissible history selection
Geometry / gravity	diffeomorphism / curvature quotient	multiplicity of admissible geometric representatives compatible with exposed curvature or boundary data

8.7 Alpha-action equivalence

Definition 8.11 (Alpha-action equivalence)**.** Let $\mathcal{F}_\alpha$ be the class of admissible history functionals $F : \mathcal{R}([x]_\Omega) \to V_c$ into a local chart value carrier V_c. Set $F \sim_\alpha G$ if F and G induce the same admissible repair selection after quotienting by the relevant chart symmetry. The parent alpha-action is the class $A_\alpha = [F]_{\sim_\alpha}$.

Remark 8.12 (Action-identity chain as a verification obligation)**.** The source wrote the chain

$$A_\alpha = [\text{path length}] = [\text{phase accumulation}] = [\text{repair effort}] = [\text{history cost}] = [\text{reconstruction order}],$$

"provided those representatives induce the same admissible selection in their chart." That proviso makes each "=" a *conditional, per-chart claim*, not an established identity: the assertion is that, *in a given chart and after its symmetry quotient*, two representatives select the same admissible repair. Each equality is therefore a verification obligation to be discharged chart by chart, and is recorded here as such rather than as a proven identity.

[Correction E17: the action-identity chain (source Definition 11) and the action-identity displays of §8 are conditional equivalences contingent on "same admissible selection per chart." They are marked as verification obligations: the local representatives below are candidate realizations, and the claim that they coincide under $\sim_\alpha$ must be checked in each chart, not assumed.]

Local representatives (each a per-chart candidate, cf. Remark 8.12): $\int L\,dt$ (classical action chart), $e^{iS/\hbar}$ (quantum phase-history chart), geodesic/path cost (geometric chart), $E-TS$ (thermodynamic free-energy chart), and repair rank/order (primitive DA chart).

8.8 Action–entropy selection

The parent object is typed as a pair $\Sigma_\Omega(r) = (A_\alpha(r), S_\Omega(r))$. A local chart supplies a scalar or ordering map $\Phi_c : (A_\alpha, S_\Omega) \to V_c$ and selects admissible repairs by $r_{\text{phys}} \in \text{Select}_{r\in\mathcal{R}([x]_\Omega)} \Phi_c(A_\alpha(r), S_\Omega(r))$.

Postulate 8.13 (Action–entropy repair selection)**.** Physical realization is selected from admissible repairs by a chart-realized action–entropy compound. The compound is parent-level; the scalar expression is chart-dependent. A local theory chooses the symmetry class G_c, measure μ_c, action representative, and selector Φ_c; it realizes the parent repair-selection structure in a stable chart rather than creating physics from nothing.

8.9 Quantum mechanics as polarity-stabilized alpha-history

In the DA reading, QM is the chart in which admissible α-histories carry oriented amplitudes and visible events require polarity-stable quantities. This matches the general fact that Born's rule maps amplitudes to probabilities while foundations still debate which assumptions *derive* the rule rather than postulate it [6,7].

Definition 8.14 (Quantum chart)**.** A quantum chart is a local realization in which admissible repairs are represented by oriented amplitudes a_i, legal transformations preserve the amplitude carrier, and visible events are polarity-invariant.

Proposition 8.15 (Polarity–Born route, conditional)**.** *Let a finite set of mutually exclusive visible alternatives be indexed by i, with oriented α-history amplitudes $a_i \in \mathbb{C}$, and suppose a visible weight w_i is (i) invariant under clock-polarity reversal $a_i \mapsto a_i^*$ and (ii) a function of the amplitude alone. Then w_i depends on a_i only through $|a_i|$, and if in addition w is (iii) multiplicative under composition of independent subsystems, $w(ab) = w(a)w(b)$, then $w_i = |a_i|^{2c}$ for some constant $c > 0$. The Born exponent $c = 1$, giving $w_i = a_i^* a_i = |a_i|^2$ and $P_i = |a_i|^2/\sum_j |a_j|^2$, follows* only *once a normalization/additivity axiom is imposed—additivity of w over mutually exclusive (orthogonal) alternatives together with the subsystem-composition law—which fixes the scale of the exponent.*

Proof. By (i)–(ii), $w_i = f(a_i)$ with $f(a) = f(a^*)$, so f factors through the polarity-invariant $|a|$; write $f(a) = h(|a|)$. By (iii), h is multiplicative on $\mathbb{R}_{\geq 0}$, hence (measurably) $h(t) = t^s$ for some $s > 0$; set $s = 2c$. The value of c is unconstrained by polarity invariance and multiplicativity alone: every $|a_i|^{2c}$ satisfies (i)–(iii). Imposing additivity over exclusive alternatives compatibly with the subsystem product pins $s = 2$ (the unique exponent under which orthogonal additivity and tensor multiplicativity are simultaneously consistent with normalization), giving $c = 1$ and the stated P_i. □

[Correction E14: the source stated this as "Lemma 1" concluding $w_i = |a_i|^2$ from polarity invariance plus "minimal nontrivial positive multiplicative invariant additive over exclusive alternatives." But polarity invariance and multiplicativity force only $w_i = |a_i|^{2c}$; any $c > 0$ is polarity-invariant, so the exponent 2 is not forced by the stated hypotheses. The real content—and the genuinely missing axiom—is the composition law that fixes the scale: additivity over orthogonal alternatives together with multiplicativity under independent subsystems (a Gleason-/homomorphism-type condition). It is therefore recast as a conditional route (Proposition), with the scale-fixing axiom named explicitly, not asserted as a lemma.]

Open proof obligation. DA must still derive the complex polarity structure itself, not merely assume the amplitude carrier is $\mathbb{C}$. The route above shows how the Born form follows *once polarity and the composition axiom are available.*

8.10 General relativity as quotient-stabilized theta-curvature

In the DA reading, GR is the chart in which D_θ exposes curvature/interval data and the stable class is taken modulo diffeomorphism and admissible geometric reconstruction; the representative may be a Lorentzian manifold with metric, connection, curvature, and stress-energy dynamics, obtained by quotient-stabilized curvature data rather than by naming a metric.

Definition 8.16 (Geometric chart)**.** A geometric chart is a local realization $c_g = (\Omega_g, G_g, \mu_g, \Phi_g)$ where G_g contains diffeomorphism/coordinate-change symmetry, D_θ exposes interval/curvature/holonomy/boundary data, and I_α supplies admissible geometric histories or variational reconstructions.

The target quotient is $[g]_\Omega = \{g' : O(g') = O(g)\ \forall O \in \Omega_g\}$, not an arbitrary metric: gauge or coordinate defects quotient out when every legal geometric observable agrees, and a non-quotiented residual becomes physical curvature, stress-energy, boundary charge, or obstruction. Recent semiclassical work uses a phase-space language in which the gravitational symplectic form and quantum Berry curvature appear in one combined structure [8]; this does not prove DA but shows the targeted seam—symplectic/geometric reconstruction together with quantum state curvature—is a live mathematical interface.

8.11 Thermodynamics as repair-multiplicity chart

Thermodynamics is the chart in which the visible state is a macrostate and the hidden sector is the class of micro-repairs compatible with it, so thermodynamic entropy is a direct realization of parent repair multiplicity, $S_{\text{Thermo}} = \log \mu_\tau(\mathcal{R}([x])/G_\tau)$, with selector $\Phi_\tau(A_\alpha, S_\Omega) \sim E - TS$ where energy and temperature are the stable observable parameters. The structural move is the same throughout: visible macro-data underdetermine admissible repair; repair multiplicity is entropy; selection balances action-like constraint against entropy-like multiplicity.

8.12 Quantum gravity as alpha-history repair over theta-curvature

The schema's reading is QG $\neq$ entropy. Entropy is residual multiplicity after quotienting; quantum gravity is the α-clocked admissible-history repair over θ-curvature data,

$$\mathrm{QG} := I_\alpha^{qg}\big(D_\theta^g X,\ K_{qg}^{\text{ret}}\big)/\!\sim,$$

where D_θ^g exposes curvature/geometric data and I_α^{qg} supplies admissible quantum-history repair. The mixed defect, when both composites are typed into a common carrier, is the clock-order residual $\Xi_{qg} = \mathrm{Res}(D_\theta^g I_\alpha^q, I_\alpha^q D_\theta^g)$ (well-typed in the sense of Definition 8.5); in the corrected reading Ξ_{qg} is not quantum gravity itself but the entropy-generating residual of incompatible quantum/geometric clock order, contributing to $S_{qg} = \log \mu(K_{qg}^{\text{ret}}/\!\sim)$.

8.13 How DA unifies the local theories

The unification is a typed statement about what each equation-class is doing, not a merger of equations.

Theory chart	DA realization	What gets quotiented
Quantum mechanics	α-history amplitude chart with polarity-stabilized visible weights	phase/gauge representatives invisible to legal measurements; non-admissible histories rejected
General relativity	θ-curvature chart with diffeomorphism-stabilized geometric reconstruction	coordinate/gauge representatives; curvature-equivalent geometric presentations
Thermodynamics	repair-multiplicity chart over macro-observables	micro-representatives compatible with the same macrostate
Classical mechanics	stationary α-action representative under low-entropy/single-repair limits	alternative histories suppressed or identified by action selection
Quantum gravity	α-history repair over θ-curvature data	quantum-geometric residuals invisible to all legal compounded operators; survivors become invariants

The formal statement is: every local physical theory is a chart c of $(X_{\mathrm{Phys}}, \Omega, \mathrm{Adm}, A_\alpha, S_\Omega, \Phi_c)$. The physical state is the terminal quotient $[x]_\Omega$; the law is the chart-realized selector Φ_c; the entropy is the measure of retained repair multiplicity; the action is the α-equivalence class of history-ordering functionals.

8.14 The terminal repair state

Proposition 8.17 (Terminal repair state)**.** *Let $(X_{\mathrm{Phys}}, \Omega, \mathrm{Adm})$ be a typed DA system whose operators compose as in Definition 8.2, with $\sim_\Omega$ defined by equality under every legal compounded operator. Then every physically visible object represented by DA is a terminal quotient state $\Psi_{\mathrm{Phys}} = [x]_\Omega$, with parent repair entropy $S_\Omega([x]) = \log \mu_\Omega(\mathcal{R}([x])/\sim_\Omega)$, and every local physical theory is a realization of this terminal state under a chart symmetry G_c, measure μ_c, and selector Φ_c.*

Proof. By definition $x \sim_\Omega x'$ exactly when all legal compounded operators return the same visible result, so no local observer using legal operations can distinguish representatives inside $[x]_\Omega$ without leakage. The admissible representatives form $\mathcal{R}([x])$; their quotient by terminal equivalence gives the retained repair multiplicity, measured by μ_Ω, whose logarithm is parent entropy. A local chart chooses G_c (identifying further representatives invisible in that chart) and a selector Φ_c over the action–entropy compound; hence the local theory is a chart-realization of the terminal quotient state. □

[Correction E15: the source labeled this "Theorem 2." Its proof unfolds the definition of $\sim_\Omega$ and the definitions of $\mathcal{R}$, μ_Ω, and parent entropy; it establishes a structural restatement rather than a new deduction, so it

is recorded as a proposition. The self-assessment of §8.15 *is retained verbatim because it is the document's honest accounting of what is and is not established.]*

8.15 What this paper does and does not prove

It does prove, internally to DA: (1) a precise terminal-quotient definition of physical state; (2) a criterion for when a defect quotients out; (3) a clock-order entropy identification of entropy with admissible repair multiplicity; (4) an alpha-action equivalence class unifying path cost, phase, repair effort, and history order; (5) a chart-relative action–entropy selection schema; (6) a conditional Born-rule route from polarity invariance.

It does not yet prove: (1) that complex Hilbert space is forced by DA alone; (2) that the Born rule follows without any additional polarity assumption; (3) that Lorentzian manifolds and Einstein dynamics are forced by DA alone; (4) that a specific new empirical quantum-gravity prediction follows; (5) that DA is more than a strong organizing language.

The strongest honest claim is therefore: DA gives a candidate parent reconstruction-selection grammar in which physics unifies as terminal admissible repair. The next task is to convert chart realizations into derivations: derive complex polarity; derive a Hilbert or algebraic quantum state space; derive Lorentzian curvature reconstruction; derive the classical Einstein limit; and identify a mixed residual not removable by gauge, coordinate choice, or renormalization convention.

8.16 Falsifiers and review obligations

DA fails as a physics-unification framework if any of the following hold.

1. **Arbitrary kernel.** K_{ret} can be expanded to save any failure without rejecting nearby inadmissible reconstructions.
2. **Gauge-only residual.** Every proposed quantum-geometric residual is removable by coordinate change, gauge choice, or known renormalization ambiguity.
3. **Imported equations.** QM, GR, or thermodynamics are inserted as postulates rather than recovered as chart realizations.
4. **No measure.** μ_Ω (and $\mu_{\theta\alpha}$) cannot be defined without smuggling in the answer.
5. **No selection rule.** Φ_c cannot reproduce any known limiting equation.
6. **No novel constraint.** The framework cannot forbid, require, or correct anything beyond existing formalisms.

The first meaningful external test is a derivation test, not a Planck-scale prediction: can the polarity–Born route be upgraded into a Born theorem without importing Hilbert structure (i.e. can the scale-fixing axiom of Proposition 8.15 be *derived*)? A second is geometric: can quotient-stabilized θ-curvature force Lorentzian signature and diffeomorphism invariance? A third is mixed: can a quantum-geometric residual be shown to survive all admissible quotienting?

8.17 Conclusion

DA unifies physics here in one specific sense: it adds no strings, loops, hidden variables, or new substance, but defines a parent carrier of admissible reconstructions; assigns θ to curvature/projection and α to history/reconstruction; treats time as their compound; identifies entropy as repair multiplicity; treats action as an alpha-history equivalence class; and selects realization by a chart-dependent action–entropy compound. The compression is: physics = terminal admissible repair selected by action–entropy under compounded clocks; with QM the polarity-stabilized alpha-history chart, GR the quotient-stabilized theta-curvature chart, thermodynamics the repair-multiplicity chart, and QG alpha-history repair over theta-curvature data. A defect quotients out when every legal compounded operator sees the same terminal state; if it does not, it is physical structure or a failed reconstruction, not a hidden mistake.

References for this chapter

[1] N. M. Kleban, *Defective Analysis repository*, 2026.

[2] N. M. Kleban & Aleph, *Framing, Terminology, and Problem Description* (Chapter 1).

[3] N. M. Kleban & Aleph, *Defective Calculus* (Chapter 2).

[4] N. M. Kleban & Aleph, *Global Definitions, Operator Cores, and Repository Discipline* (Chapter 7).

[5] N. M. Kleban & Aleph, *Witnessed Decomposition Systems: Formal Definition Draft*, 2026.

[6] P. Östborn, "Born's rule from epistemic assumptions," arXiv:2402.17066, 2024.

[7] S. S. Afonin, "The Born rule for quantum probabilities from Newton's third law," arXiv:2408.03941, 2024.

[8] A. Bhattacharya & O. Parrikar, "Covariant phase space and the semi-classical Einstein equation," arXiv:2510.19939, 2025.

[9] B. A. Juárez-Aubry, "Recent developments in semiclassical gravity," arXiv:2509.02051, 2025.

[10] J. Aziz & R. Howl, "Classical theories of gravity produce entanglement," Nature **646**, 813–817, 2025. *(See the footnote in §1: this result is contested by subsequent re-analyses.)*

[11] S. Bose et al., "A Spin-Based Pathway to Testing the Quantum Nature of Gravity," arXiv:2509.01586, 2025.

[12] F. Marchesano, G. Shiu, & T. Weigand, "The Standard Model from String Theory: What Have We Learned?" arXiv:2401.01939, 2024.

Appendix A

Corrections Register

This appendix records, with a one-line rationale each, the corrections folded into the consolidated edition. References point to the chapter where each is applied. The register is exhaustive for the substantive corrections; purely cosmetic reformatting (display reflow, symbol spacing) is not itemized.

Per-document corrections

E1 (Ch. 2, no-leakage).
The source's no-leakage statement $k \sim_{\mathrm{hid}} k' \Rightarrow \Pi(k) = \Pi(k')$ is a tautology under $K_{\mathrm{hid}} = \ker \Pi$. Restated as a constraint on downstream visible composites: hidden-equivalent retained data must yield reconstructions no legal visible map can separate.

E2 (Ch. 2, polarity).
The classifier σ had three incompatible domains. Fixed to a single signature $\sigma : Y \to \Sigma \cup \{\perp\}$, with an explicit extended form $\sigma(y, \rho)$ for the operational regime.

E3 (Ch. 2, residual).
The clock residual is written $\Delta_{\theta,\alpha}(X; k)$ to expose its dependence on the retained datum k, with both compared terms in Y so the comparison is well-typed.

E4 (Ch. 2, symbols).
Probe power, integer cutoff, and retained-data class were all denoted with colliding letters. Renamed: probe step m, cutoff M, class K.

E5 (Ch. 3, fiber vs kernel).
The source set $K_{\mathrm{ret}}([G]) = \delta_n^{-1}(\mathrm{Deck}(G))$, conflating fiber and kernel. Corrected per decision (D1): the deck fiber carries the candidate class; the retained kernel is the selection datum within it. The reconstruction conjecture is the fiber-singleton question.

E6 (Ch. 6, projected interference).
The projected-norm formula dropped the amplitude conjugate. Corrected to $\overline{A_k} A_l e^{i(\psi_l - \psi_k)}$ with a declared inner-product convention (conjugate-linear first slot); the source's form is recovered exactly in the polar gauge $A_k \geq 0$.

E7 (Ch. 6, phase residual).
The finite phase residual is t-independent only on frequency-matched pairs $\omega_{ik} = \omega_{jl}$; the general relative phase carries a $(\omega_{ik} - \omega_{jl})t$ term, now exposed.

E8 (Ch. 6, notation/status).
The atlas, a transport operator, and tensor amplitudes shared the symbol A; separated into $\mathcal{A}$, A,

A_k. The "finite latent magnitude invariance" statement, an immediate consequence of $|e^{i\psi}| = 1$, is demoted from Theorem to Remark.

E9 (Ch. 7, reconstruction equivalence).
The source relation used unquantified existential witnesses and is not transitive. Redefined as its transitive closure (equivalently the section/quotient realization, which is transitive).

E10 (Ch. 7; carried to Ch. 8, Ch. 1).
The clock-order residual written directly as $D_\theta I_\alpha$ vs $I_\alpha D_\theta$ is ill-typed (Y versus $X/\sim$). The well-typed primary object is $\Delta_{\theta,\alpha}(X;k)$ with both terms in Y; the commutator is a secondary shorthand contingent on a declared comparison functor and a retained datum.

E11 (Ch. 7, containment).
Recon $\subsetneq$ DC $\subsetneq$ DA records how three vocabularies nest; nothing is deduced. Demoted from Proposition to Remark.

E12 (Ch. 5, metric).
"Metric geometry requires $\mathbb{R}$" is too strong (non-Archimedean, rational-valued, lattice-valued metrics exist). Softened: limit-stable comparison needs a *complete value carrier*; for the generator-visible Archimedean size that carrier is $\mathbb{R}$, but the requirement is completeness of the value object.

E13 (Ch. 8, clock-order entropy).
The entropy "theorem" is the definition of $S_{\theta\alpha}$; its deductive content is the vanishing-residual criterion (singleton repair class $\Leftrightarrow$ zero entropy). Split into Definition + Proposition; the measure $\mu_{\theta\alpha}$ is declared with a standing construction obligation.

E14 (Ch. 8, polarity–Born).
Polarity invariance plus multiplicativity force only $w_i = |a_i|^{2c}$; the exponent 2 is not pinned by the stated hypotheses. Recast as a conditional route (Proposition) and the scale-fixing axiom (orthogonal additivity together with subsystem multiplicativity) named explicitly.

E15 (Ch. 8, terminal state).
The terminal-repair "theorem" is a structural restatement of the definition of $\sim_\Omega$ and parent entropy. Demoted to Proposition; the §"what this paper does/does not prove" self-assessment kept verbatim.

E16 (Ch. 8, operator system).
Composition of legal operators is partial, so the generated structure is a typed partial monoid (equivalently the morphisms of a category of admissible typed carriers), not a monoid.

E17 (Ch. 8, action identities).
The action-identity chain $A_\alpha = [\text{path length}] = \cdots$ is conditional on "same admissible selection per chart." Each equality is marked as a per-chart verification obligation, not an asserted identity.

Cross-document harmonization

Notation $\oplus$ vs $\veebar$.
The framing document used $\oplus$ for the local/global disjunction (exclusive-or), while the COTG and addendum documents used $\oplus$ for a direct sum of sectors. Split into $\veebar$ (exclusive-or, framing diagnostic) and $\oplus$ (direct sum, sector decompositions); both are used with their correct meaning throughout.

Malformed as an assignment property.
"Malformed" is a predicate on the clock *assignment* ($\mathrm{Clk}_{\mathrm{assigned}} \neq \mathrm{Clk}_{\mathrm{true}}$), not a third value of $\mathrm{Clk}(P) \subseteq \{\theta, \alpha\}$. The framing chapter reports a well-formed signature plus a separate well-assigned/malformed verdict, keeping the types separate.

Projection-domain drift.
The visible quotient/comparison map Π was applied sometimes to retained data and sometimes to states. Its domain is fixed per instance at the point of use and stated explicitly where it acts.

Clock-index doctrine (D4).

The symbols θ, α denote operation clocks—indices over operation families—uniformly in the spine, the addendum's "clock coordinates," and the scalar ladder, where they index the successor/aggregation operations. The word "coordinate" in "clock coordinate" abbreviates operation-index axis (the clock sense) and is distinguished from a geometric coordinate (dimension) and from ordinary time, per the editorial preface. The scalar-ladder bridge is stated in §4.1.

Citation status (Ch. 8).

References [6], [10], [11] were verified against primary sources (arXiv:2402.17066; Nature **646**:813–817, DOI 10.1038/s41586-025-09595-7; arXiv:2509.01586). Reference [10] is flagged as contested by subsequent re-analyses; a neutral footnote records this and notes that the dispute strengthens the "open problem" framing. References [7], [8], [9], [12] are retained as given.